

EBERHARD KARLS UNIVERSITÄT TÜBINGEN  
Prof Dr. Hannah Markwig

Bachelor thesis

# Quartics, Cubics and bitangents

27 Lines on Cubics and 28 Lines on Quartics

Handed in: May 11, 2022  
by David Klein

Adresse:  
Alberstraße 36  
72074 Tübingen

david.klein@student.uni-tuebingen.de  
Bachelor of Education  
Mathematik / Evangelische Theologie  
Bachelor of Science  
Mathematik

---

## Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                             | <b>1</b>  |
| <b>2</b> | <b>Preliminaries</b>                            | <b>2</b>  |
| <b>3</b> | <b>The 27 lines on a cubic</b>                  | <b>6</b>  |
| <b>4</b> | <b>Generalising results to different Fields</b> | <b>12</b> |
| <b>5</b> | <b>The 28 bitangents</b>                        | <b>20</b> |
| <b>6</b> | <b>The Shadow of a Cubic</b>                    | <b>28</b> |
| <b>7</b> | <b>Conclusion and outlook</b>                   | <b>35</b> |
| <b>8</b> | <b>Notation</b>                                 | <b>36</b> |
|          | <b>References</b>                               | <b>37</b> |
|          | <b>Source code</b>                              | <b>38</b> |

## 1 Introduction

The study of algebraic geometry is an old art in the area of pure mathematics. Whilst algebraic curves can be studied with analytical methods and the quadrics can be studied through linear algebra, cubics are the easiest form that really need the tools of algebraic geometry. In the bigger theme of looking for structure in the category of algebraic varieties, the find of a 1 to 1 correspondence between two classes is a helpful construct. Moreover, if that correspondence comes with an easy to see geometric interpretation, this becomes a beautiful finding. Topic wise, this finding is the leading theme of the thesis. We want to understand the 1 to 1 correspondence between smooth cubics in 3 – *space* and smooth quartics in the plane. To give that correspondence a geometric interpretation, we will first prove that there are 27 lines contained in a smooth cubic, and that there are 28 bitangents at a smooth quartic. Later, it will turn out that the shadow of the 27 lines will be the bitangents of the quartic. These theorems will hold not just for varieties as zero sets of polynomials over the complex numbers, but in much broader settings. To understand that setting and make us familiar with the language spoken in algebraic geometry, we will get to know the spaces called scheme. Thus, we have 4 topics in this thesis:

1. There are 27 lines on a smooth cubic.
2. Discovering the world of schemes.
3. There are 28 bitangents on a smooth plane quartic.
4. The shadow of a smooth cubic.

In addition to the beauty of the results, there is another reason for those topics. In recent times, the idea of throwing a *piece wise linear shadow*, called a tropicalization, turned out to be a useful tool for studying the algebraic varieties. In fact, the study of such piece wise linear varieties called tropical varieties became a topic of its own, where there is still much to be discovered. For a tropical quartic, there are 7 classes of bitangents with 4 special cases that can be lifted( [LM20]). So there are in a sense 28 bitangents. The study of lines in a cubic is a more subtle, as seen in [Gei19]. Now, it is a natural question, whether there is a connection between lines in the cubic and the tropical bitangents. In order to find answers to that question, we want to get to know the shadow of algebraic cubics to quartics. To do that, we proved the properties of the shadow computationally and developed an algorithm to construct such a shadow. This idea can be seen in figure 1. The second chapter's importance arises from the observation that we need our results and algorithms to hold over arbitrary algebraic fields, since we can tropicalize varieties over valuation fields such as the Puiseux series. Whilst this idea dictates our topics, there is another choice to be made. That is how much one proves, and which proofs one uses, if there are several ones. The chosen ideal by which the proves need to be justified is the ideal of beauty, which by no means is easy to define nor to archive. However, as an idea, let's say a proof that connects different areas of

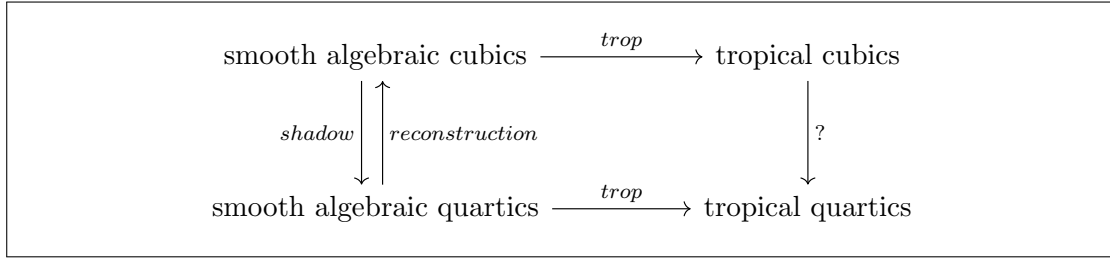


Figure 1: Tropical shadow

mathematics is rather unexpected and insightful, or one that is general and geometrically easy to follow. These 2 conditions, the themes and the ideal, dictate the following thesis.

## 2 Preliminaries

In this chapter, we will first introduce algebraic geometry, by looking at embedded algebraic surfaces. We review the important theorems and motivate them without any proofs. The aim is that the reader gets used to the notation and has the definitions in mind that we use in this thesis. We will assume knowledge in the field of commutative algebra such that we can use tools like quotient rings, localizations, morphisms or functors.

### Algebraic varieties

Algebraic geometry is a branch of mathematics that studies geometric objects coming from algebraic definitions. With that in mind, we define such a geometric object as the set of points that satisfy a polynomial relation. If we talk about varieties in  $\mathbb{R}^n$  we look at a polynomial  $f$  in  $n$  variables and get the variety by considering all points  $x$  in  $\mathbb{R}^n$  that satisfy that  $f(x) = 0$ . Since our  $x$  also satisfy  $f^2(x) = 0$  and  $f^3(x) + p(x) \cdot f(x) = 0$ , for a polynomial  $p$ , we can widen our definition to the following:

**Definition 1** (Algebraic variety). For an ideal  $I$  living in the polynomial ring  $K[x_1, \dots, x_n]$ , where  $K$  is a Field, we define the variety of  $I$  as the zero-set of all the polynomials in  $I$ , or as

$$V(I) = \{x \in K^n \mid f(x) = 0 \ \forall f \in I\}.$$

However one notices, that not all ideals lead to different varieties. For example, if we look at  $I := \langle x^2 + y^2 + 1 \rangle$  as an ideal in  $\mathbb{R}[x, y]$  we realize that  $V(I) = \emptyset = V(R[x, y])$ . The reason for that is, that  $\mathbb{R}$  is not algebraically closed and if we require our field to be just that our problem almost disappears. The last problem that can appear is that some ideals contain powers of polynomials whilst not containing the polynomials itself. Since the roots of a polynomial don't change by exponentiation, we have to consider

ideals that contain every polynomial that is contained as a higher power. If we do that, by Hilbert's Nullstellensatz we have 1 to 1 correspondence between radical ideals and algebraic varieties. From now on, all varieties will be over an algebraically closed field.

**Remark 1** (Intersections and unions of varieties). For a family,  $V(I_\lambda)$  ( $\lambda \in \Lambda$ ), we can calculate the arbitrary unions and the finite intersections as follows:

$$\begin{aligned} \bigcup_{\lambda \in \Lambda} V(I_\lambda) &= V\left(\bigcap_{\lambda \in \Lambda} I_\lambda\right) \\ \bigcap_{\substack{\lambda \in U \subset \Lambda \\ \#U < \infty}} V(I_\lambda) &= V\left(\bigcup_{\substack{\lambda \in U \subset \Lambda \\ \#U < \infty}} I_\lambda\right). \end{aligned}$$

This intersection reversion is a common theme which will be all over algebraic geometry. In fact, when algebraic varieties are defined as schemes, we have a contravariant functor between rings and schemes.

In this work we mostly consider smooth hypersurfaces which are algebraic varieties defined by only one polynomial or a principle ideal. The notion of smoothness follows our intuition of smooth meaning that there isn't an abrupt change of direction, which would yield a tangent space that is of higher dimension. That observation is captured by the following definition of smoothness.

**Definition 2** (Smooth algebraic variety). A algebraic variety  $X$  defined by the ideal  $I = \langle f_1, \dots, f_r \rangle$  in  $K^n$ , where  $K$  is an algebraically closed Field is smooth iff for all points  $p$  in  $X$  the formal Jacobian-Matrix

$$\left( \frac{\partial f_i}{\partial x_j}(p) \right)_{i,j}$$

has rang  $n - \dim(X)$ . Since we only consider hypersurfaces, that are of co-dimension 1, our Matrix needs the rank 1.

For those affine varieties, we realize that they might coincide with how we think of reality but not how we see reality. When we look at our World, we consider lines pointing directly to our eye as points. So, if we imagine our eyes being the origin in 3-Space, we can see everything up to multiples. Following that projective idea, we construct the protective space over the complex field as the quotient of  $(n+1)$ -space without the zero and  $\mathbb{C}^*$ .

**Definition 3** (Projective space).

$$\mathbb{P}^n = \mathbb{C}^{n+1} \setminus \{0\} / \mathbb{C}^*$$

We will write points in  $\mathbb{P}^n$  as  $[x_0 : x_1 : \dots : x_n]$  to emphasize that we are interested in relations instead of absolute values. To define varieties in the projective realm, we need

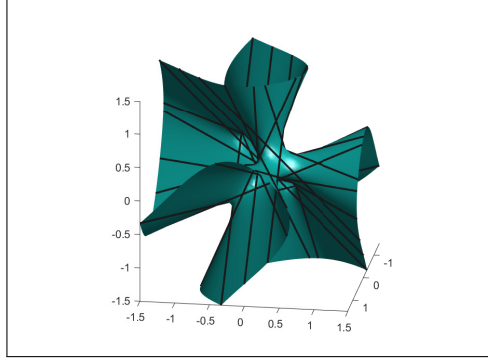


Figure 2: Smooth Cubic

to be careful to archive well-defined zero sets of polynomials  $P$ . Because now have to ensure, that if  $P(x) = 0$  then also  $P(\lambda x) = 0$  for  $\lambda$  in  $K$ . For that reason, we need to consider ideals, that are generated by homogeneous polynomials, since for those the zero set is closed under scalar multiplication. We call those ideals homogeneous.

**Definition 4** (Projective variety). For a given homogeneous ideal  $I$  in  $K[x_0, \dots, x_n]$  we set

$$V(I) = \{x \in \mathbb{P}^n \mid f(x) = 0 \text{ for all } f \text{ in } I\}.$$

If we restrict ourselves now to projective hypersurfaces defined by the polynomial  $f$  and apply the Jacobian criteria, we end with the following:  $V(I)$  is smooth iff  $\left(\frac{\partial f}{\partial x_i}(p)\right)_{i=0 \dots n}$  only vanishes in  $p = 0$ .

**Remark 2.** Those definitions are sufficient for now. However, we will later in definition 7 expand our view of algebraic varieties to so-called schemes, that can be seen as locally affine varieties glued together.

**Example 1.** We can consider the variety  $V(x_0^2x_2 + 2x_0^2x_3 - 2x_0x_1x_2 - x_0x_1x_3 - x_0x_2^2 + x_1^2x_3 + x_2^2x_3 - x_0x_3^2)$  as a subset of  $\mathbb{P}^3$ . To visualize it, we have to go to affine charts. That means to cut out a hyperplane like  $x_3 = 0$ . By doing that in our variety and in  $\mathbb{P}^3$ ,  $x_3$  cannot vanish and since we are only interested in the coordinates up to multiples we can assume  $x_3 = 1$  and by doing that fix our coordinates to affine ones. If we do that, we get the picture shown in figure 2. This picture already hints a remarkable fact, that there are lines contained in the cubic.

**Definition 5** (Zariski topology). For a given affine algebraic variety,  $V(I)$  we define all varieties contained in  $V$  as closed subsets. Since we can unite arbitrary families and intersect finite, we have a well-defined topology. Since we can unite ideals, this topology restricted to a subset is equal to the subset topology. This construction of a topology works analogously for projective varieties and going into affine charts of a projective

variety is nothing but looking at the variety on an open subset  $\mathbb{P}^n \setminus V(x_i)$ . We call a closed subset of  $V(I)$  irreducible if it is given by a prime ideal, and we can notice, that the irreducible subsets are in 1 to 1 correspondence with the prime ideals containing  $I$ .

**Definition 6** (Dimension). We define the dimension of an algebraic variety  $X = V(I)$  to be the Krull dimension of the topological space, meaning the maximal length of a chain of irreducible closed contained subsets. Since  $K[x_0, \dots, x_n]$  is Noetherian, we have a Noetherian topological space, where all decreasing chains of closed contained subsets get stationary. The topological Krull dimension is then the same as the Krull dimension of the so-called *coordinate ring*  $K[X] = K[x_0, \dots, x_n]/I$ , which is proven once we realize that the prime ideals that contain  $I$  are the prime ideals in  $K[X]$ .

**Example 2.** The coordinate ring of  $K^n = V(\langle 0 \rangle)$  is  $K[x_0, \dots, x_n]$ .

**Remark 3** (Maximal ideals are points). For any given coordinate ring  $K[X]$ , there is a 1 to 1 correspondence between points in  $X$  and maximal ideals in  $K[X]$ .

*Proof.* One direction is clear, since any point  $p = (p_1, \dots, p_n)$  comes from the ideal  $\mathfrak{m}_p = \langle x_1 - p_1, \dots, x_n - p_n \rangle$ , and there is an obvious isomorphism such that

$$K[x_0, \dots, x_n]/\mathfrak{m}_p \cong K.$$

The other direction is more tricky, and we will just sketch the idea. One can show that  $K \subset L := K[X]/\mathfrak{m}$  is a finite field extension, since it is a finite  $K$  algebra. This however means that  $K = L$  since  $K$  is algebraically closed, giving us that the canonical map  $\phi : K \rightarrow K[X]/\mathfrak{m}$  is an isomorphism. If we define  $p_i = \phi^{-1}([x_i])$ , we have

$$[x_i] = \phi(\phi^{-1}([x_i])) = \phi(p_i) = [p_i].$$

That is why  $[x_i - p_i] = 0$  meaning  $x_i - p_i \in \mathfrak{m}$ , and by the maximality of  $\mathfrak{m}$  and of  $\langle x_1 - p_1, \dots, x_n - p_n \rangle$  we have their equality.  $\square$

This is why we can encode the topological space in the coordinate ring, since  $X \cong \text{spec}_{\mathfrak{m}}(K[X])$  the set of all maximal ideals. This idea is what will later motivate the construction of schemes.

Lastly, we want to get to know a translation from the algebraic to the analytical world. For that have the implicit function theorem:

**Theorem 1** (Implicit function theorem). *Let  $D \subset \mathbb{R}^r \times \mathbb{R}^n$  be open, and  $f : D \rightarrow \mathbb{R}^n, (u, x) \mapsto f(u, x)$  be continuously differentiable. Let  $(a, b)$  be a point in  $D$ , such that  $f(a, b) = 0$  and  $\det \frac{\partial f}{\partial x}(a, b) \neq 0$ .*

*Then there exist open neighbourhoods  $U$  of  $a$  and  $V$  of  $b$  and a continuously differentiable function  $\phi : U \rightarrow V$  with  $\phi(a) = b$ , such that*

$$f(u, x) = 0 \quad \Leftrightarrow \quad x = \phi(u).$$

*In other words, the curve  $V(f) \subset D$  is locally a graph  $\{(u, \phi(u))\}$  if the Jacobian of in  $x$  doesn't vanish.*

### 3 The 27 lines on a cubic

It is a historical and remarkable fact that any smooth variety of degree 3 in  $\mathbb{P}^3$  contains exactly 27 lines. We call those varieties cubics, and a line is a variety defined by two homogeneous polynomials of degree 1. We will prove this fact by introducing the concept of spaces of  $k$ -dimensional subspaces. This way we will get a deep understanding of lines and their configurations and end up with a proof that uses algebraic and analytical methods. For this whole chapter, we will work over the complex numbers and loosely follow Andreas Gathmann's chapter on 27 lines on a cubic [Gat21a].

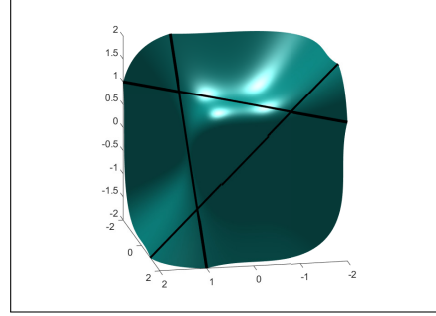


Figure 3: Fermat cubic

**Theorem 2.** *There are exactly 27 lines on a smooth cubic in  $\mathbb{P}^3$ .*

To show this, we will use, that the Fermat cubic  $V(x_0^3 + x_1^3 + x_2^3 + x_3^3)$  has exactly 27 lines. Afterwards, we will show that all smooth cubics have the same amount of lines. To do that, we will need to view

$$\mathcal{M} = \{(X, L) | X \text{ is a smooth cubic, } L \text{ is a line in } X\}$$

as a Zariski closed subvariety of  $U \times G(2, 4)$ , where  $U$  is the set of all smooth cubics and  $G(2, 4)$  the set of lines in  $\mathbb{P}^3$ . Thus, we will need to develop an understanding of  $U \times G(2, 4)$ .

**Lemma 1.** *The Fermat cubic has 27 lines.*

*Proof.* There are lines given by the two equations

$$\alpha x_0 = x_1 \text{ and } \beta x_2 = x_3,$$

where  $\alpha$  and  $\beta$  are third roots of -1. The vanishing sets of those two polynomials are obviously contained in the Fermat cubic and since there is always 3 third roots of -1, we already found 9 lines. To find the missing 18 lines, we permute the 4 coordinates. If we take any tuple  $(i, j)$  for  $0 \leq i, j \leq 3$  to build the first equation  $\alpha x_i = x_j$ , the second equation is fixed on the remaining 2 variables, and therefore 2 chosen tuples give the same result. Since there is  $\binom{4}{2} = 6$  possible tuples, we end up with 3 possible configurations, giving us  $3 * 9 = 27$  linear subvarieties. Since -1 is a third root of -1, we get 3 lines, that lie in the real world as shown in figure 3.

□

We now want to understand  $U \times G(2, 4)$ , starting with  $U$ .

**Lemma 2** (Space of smooth cubics). *The set of all smooth cubic hypersurfaces of  $\mathbb{P}^3$  can be identified with a connected subset of  $\mathbb{P}^{19}$ , using the classical topology.*



*Proof.*  $U$  is a subset of all cubic varieties, which can be identified with the set of all homogeneous polynomials in 4 variables of degree 3 modulo scalars. Since for each monomial we have to take 3 variables out of the 4 possible ones (with returning them), we get  $\binom{4+3-1}{3} = 20$  possibilities. That is why

$$U \subset \mathbb{C}[x_0, \dots, x_3]_d / \mathbb{C} = \mathbb{C}^{20} / \mathbb{C} = \mathbb{P}^{19}.$$

□

**Remark 4** (Notation). If we consider cubic hypersurfaces, they are given by (classes of) polynomials of the form

$$f_c = \sum_{\alpha} c_{\alpha} x^{\alpha}, \quad (1)$$

where  $\alpha = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$  s.th.  $\sum_i \alpha_i = 3$ ,  $x = (x_0, x_1, x_2, x_3)$  and  $c \in \mathbb{P}^{19}$ .

**Lemma 3** (Grassmanians).  $G(2, 4)$  as the set of all lines in  $\mathbb{P}^3$  is a projective 4 dimensional variety.

*Proof.*  $G(2, 4)$  is the set of 2 dimensional subspaces of  $\mathbb{C}^4$  or the set of 1 dimensional subspaces of  $\mathbb{P}^3$ . It is itself a protective variety of dimension 4. That can be seen through the following mapping of sets:

$$\begin{aligned} \phi : \{\text{lines in } \mathbb{P}^3\} &\rightarrow \mathbb{P}^6, \\ \alpha, \beta \text{ s.th. } \text{Lin}(\alpha, \beta) = L &\mapsto \alpha \wedge \beta \end{aligned}$$

that takes from every line 2 different points  $[a_0 : a_1 : a_2 : a_3]$ ,  $[b_0 : b_1 : b_2 : b_3]$  and maps them to the point  $\alpha \wedge \beta$  in  $\mathbb{P}^6$  given by the 6 maximal minors of  $\begin{bmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \end{bmatrix}$  with a fixed order. That map is well-defined concerning its codomain because it only vanishes if all maximal minors vanish and therefore only if the rows are linearly dependent. That, however, is never the case since we chose different points on the line. Now if we take two different points  $\gamma, \delta$  on the same line, those can be obtained via 2 types of linear transformations ( $\alpha \mapsto \lambda\alpha$  and  $\alpha \mapsto \alpha + \lambda\beta$ ) s.th.  $\gamma = \lambda_1\alpha + \mu_1\beta$ , and  $\delta = \lambda_2\alpha + \mu_2\beta$ . However, through the equalities

$$\alpha \wedge \beta = (\alpha + \lambda\beta) \wedge \beta \quad \text{and} \quad \lambda(\alpha \wedge \beta) = (\lambda\alpha) \wedge \beta, \quad (2)$$

we can see that  $\phi$  is well-defined regardless of the representation of a point in  $\mathbb{P}^3$  and the choosing of representations of the Line.

That map is injective, since if  $\alpha \wedge \beta = \gamma \wedge \delta$ , then we can add  $\gamma$  on both sides to get  $\gamma \wedge \alpha \wedge \beta = \gamma \wedge \gamma \wedge \delta$ . Now each side stands for the ordered maximal minors of the  $3 \times 4$ -Matrix. The right side now computes to zero caused by its linear dependence and therefore  $\alpha, \beta, \gamma$  are linearly dependent. The same argument holds for  $\delta$  and we get that  $\text{Lin}(\alpha, \beta) = \text{Lin}(\gamma, \delta)$ . Now we know that we can embed  $G(2, 4)$  in the projective Space. That argument can be generalized for  $k$ -dimensional subspaces of  $\mathbb{P}^n$ , and is called the

Plücker embedding, where „ $\wedge$ “ stands for an alternating tensor product. The map  $\phi$  maps to the  $k$ -fold alternating tensor product of  $K(\Lambda^k K^n)$ . The dimension of

$$\Lambda^k K^n$$

equals the number of maximal minors of the  $k \times n$ -matrix minus 1 because the minors can be used as a basis. Therefore, the dimension of  $G(k, n) = \binom{n}{k} - 1$ , because to make maximal minors we choose  $k$  columns out of the  $n$ .

To see that this embedding of  $G(2, 4)$  yields a projective variety, we have to analyze the image of  $\phi$ .  $w = [w_0 : w_1 : w_2 : w_3]$  is in the image iff it is the alternating tensor product of 2 linear independent points in  $\mathbb{P}^3$ . By (2), we realized that linear transformations don't affect the outcome, which is why we can without restriction require  $w$  to have the form

$$[\lambda : 0 : \lambda\alpha_2 : \lambda\alpha_3] \wedge [0 : \lambda : \lambda\beta_2 : \lambda\beta_3] = [\lambda : \lambda\beta_2 : \lambda\beta_3 : -\lambda\alpha_2 : -\lambda\alpha_3 : \alpha_2\beta_3 - \beta_2\alpha_3].$$

However, that is the case iff  $w_0w_5 + w_2w_3 - w_1w_4 = 0$ . And therefore  $G(2, 4)$  is a projective hypersurface in  $\mathbb{P}^6$ , that is smooth, which can be checked via the Jacobian criteria.  $\square$

**Remark 5.** Instead of looking at  $G(2, 4)$  as the variety given by  $w_0w_5 + w_2w_3 - w_1w_4 = 0$ , we could have inspected the map  $f_w : K^n \rightarrow \Lambda^k K^n; v \mapsto v \wedge w$  for every  $w$  in  $\Lambda^k K^n$ . Now it can be shown that  $w$  is in  $G(k, n)$  iff  $f_w$  has rank  $n - k$  ([Gat21a, p. 8.14]), which is true iff all  $(n - k + 1) \times (n - k + 1)$ -minors vanish. Since those are polynomials, we can see in more general that  $G(k, n)$  is a projective variety. Then, however, our  $G(2, 4)$  would be the zero set of all  $3 \times 3$ -minors of a  $4 \times 4$  matrix. So we would get  $\binom{4}{3} = 4$  possible ways to choose the rows and columns, and therefore 16 cubic equations.

**Remark 6.** If we consider  $U_0$  as the space in  $G(2, 4) \subset \mathbb{P}^5$  where the first coordinate is non-zero, we effectively consider those lines, where the first two minors are non-zero. Therefore, after a transformation following (2) we can look at lines defined by the rows of  $\begin{pmatrix} 1 & 0 & a_2 & a_3 \\ 0 & 1 & b_2 & b_3 \end{pmatrix}$ . So locally, the lines are given by 4 entries. Since different  $(a_2, a_3, b_2, b_3)$  lead to different lines, and every span that is defined by points that are independent considering the first  $k$  coordinates, we get the bijection

$$\begin{aligned} f : \mathbb{A}^4 &\rightarrow G(2, k) \cap U_0 \\ (a_2, a_3, b_2, b_3) &\mapsto (1, 0, a_2, a_3) \wedge (0, 1, b_2, b_3) = [1 : b_2 : b_3 : -a_2 : -a_3 : a_2b_3 - a_3b_2]. \end{aligned} \quad (3)$$

That is a morphism since  $\wedge$  is a polynomial in every entry. Since the entries of  $f$  can be taken out of its image, the inverse of  $f$  is just a projection up to sign. Therefore,  $f$  is an isomorphism. For a deeper understanding of a morphism of this kind, read definition 7 and the proof of theorem 7.

*Proof of Theorem 2.* Since we now understand  $U$  and  $G(2, 4)$ , we can look at

$$\mathcal{M} = \{(X, L) | X \text{ is a smooth cubic, } L \text{ is a line in } X\} \subset U \times G(2, 4)$$

as a subset of an algebraic variety. Using the local coordinates of (3) and the notation for cubics from (1), we can see:

$$\begin{aligned} (c, a_2, a_3, b_2, b_3) \in \mathcal{M} &\Leftrightarrow f_c(s(1, 0, a_2, a_3) + t(0, 1, b_2, b_3)) = 0 \quad \forall s, t \in K \\ &\Leftrightarrow \sum_{\alpha} c_{\alpha} (s(1, 0, a_2, a_3) + t(0, 1, b_2, b_3))^{\alpha} = 0 \quad \forall s, t \\ &\Leftrightarrow \sum_{\alpha} c_{\alpha} s^{\alpha_0} t^{\alpha_1} (sa_2 + tb_2)^{\alpha_2} (sa_3 + tb_3)^{\alpha_3} = 0 \quad \forall s, t \end{aligned}$$

In the last sum, every monomial falls into 1 of the following categories:

1.  $c_{\alpha} s^{\alpha_0} t^{\alpha_1} s^{\alpha_2} a_2^{\alpha_2} s^{\alpha_3} a_3^{\alpha_3} = s^{\alpha_0 + \alpha_2 + \alpha_3} t^{\alpha_1} c_{\alpha} a_2^{\alpha_2} a_3^{\alpha_3} = s^i t^{3-i} c_{\alpha} a_2^{\alpha_2} a_3^{\alpha_3}$
2.  $c_{\alpha} s^{\alpha_0} t^{\alpha_1} s^{\alpha_2} a_2^{\alpha_2} t^{\alpha_3} b_3^{\alpha_3} = s^i t^{3-i} c_{\alpha} a_2^{\alpha_2} b_3^{\alpha_3}$
3.  $c_{\alpha} s^{\alpha_0} t^{\alpha_1} t^{\alpha_2} b_2^{\alpha_2} t^{\alpha_3} b_3^{\alpha_3} = s^i t^{3-i} c_{\alpha} b_2^{\alpha_2} b_3^{\alpha_3}$
4.  $c_{\alpha} s^{\alpha_0} t^{\alpha_1} t^{\alpha_2} b_2^{\alpha_2} t^{\alpha_3} a_3^{\alpha_3} = s^i t^{3-i} c_{\alpha} b_2^{\alpha_2} a_3^{\alpha_3}$

Thus, we can sort the monomials and rewrite our condition to

$$\begin{aligned} (c, a_2, a_3, b_2, b_3) \in \mathcal{M} &\Leftrightarrow \sum_{i=0}^3 s^i t^{3-i} F_i(c, a_2, a_3, b_2, b_3) = 0 \quad \forall s, t \in K \\ &\Leftrightarrow s^i t^{3-i} F_i(c, a_2, a_3, b_2, b_3) = 0 \quad 0 \leq i \leq 3. \end{aligned}$$

Therefore,  $\mathcal{M}$  is closed in the Zarisky topology of  $U \times G(2, 4)$ . Since  $f_c(s(1, 0, a_2, a_3) + t(0, 1, b_2, b_3)) = 0 \Leftrightarrow s^i t^{3-i} F_i(c, a_2, a_3, b_2, b_3) = 0$ , we can from now on interpret  $f_c$  as a function taking  $(c, a_2, a_3, b_2, b_3)$  and mapping to  $\mathbb{C}^4$ .

Next, we want to show that  $\mathcal{M}$  is locally the graph of a continuously and differentiable function  $U \rightarrow G(2, 4)$ , regarding the classical topology on  $\mathbb{C}$ . Because then the amount of lines cannot change, by locally changing the smooth cubic. Since we want to switch to an analytical world, we now use the Euclidean topology for the rest of the argument.

To show that, we will use the implicit function theorem, which is why we need to show that the Jacobian never vanishes in the latter coordinates. To get there, we view  $\mathcal{M}$  as the set satisfying  $f_c(s(1, 0, a_2, a_3) + t(0, 1, b_2, b_3)) = \sum_{i=0}^3 s^i t^{3-i} F_i(c, a_2, a_3, b_2, b_3) = (0, 0, 0, 0)$  and see, that  $\left(\frac{\partial(\sum_i s^i t^{3-i} F_i)}{\partial(a_2, a_3, b_2, b_3)}\right)$  is invertible in  $(a_2, a_3, b_2, b_3)$  which we will transform to be  $(0, 0, 0, 0)$ . We can now use the chain rule to get:

$$\begin{aligned} &\frac{\partial}{\partial a_2} \sum_i s^i t^{3-i} F_i(c, a_2, a_3, b_2, b_3) \Big|_{(c, a_2, a_3, b_2, b_3) = (c, 0)} \\ &= \frac{\partial}{\partial a_2} f_c(s(1, 0, a_2, a_3) + t(0, 1, b_2, b_3)) \Big|_{(c, 0)} \\ &= \frac{\partial}{\partial a_2} f_c((s, t, sa_2 + tb_2, sa_3 + tb_3)) \Big|_{(c, 0)} \\ &= s \frac{\partial}{\partial x_2} f_c(s, t, 0, 0). \end{aligned}$$

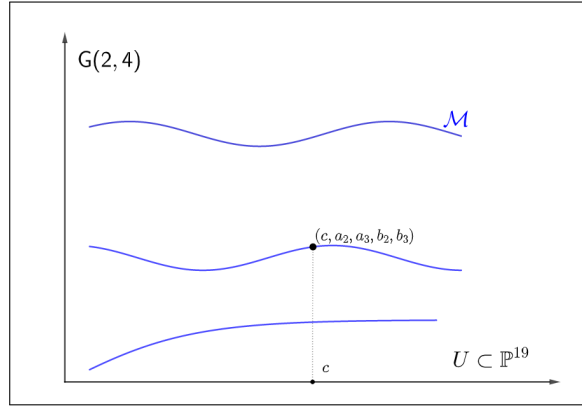


Figure 4: Locally a Graph

The results for the other coordinates are analogous, and we get the Jacobian matrix

$$\left( s \frac{\partial}{\partial x_2} f_c(s, t, 0, 0), s \frac{\partial}{\partial x_3} f_c(s, t, 0, 0), t \frac{\partial}{\partial x_2} f_c(s, t, 0, 0), t \frac{\partial}{\partial x_3} f_c(s, t, 0, 0) \right).$$

Now, let's assume that the Jacobian wasn't invertible. Then, the columns would be linearly dependent or in other words, there would exist  $(\lambda_2, \lambda_3, \mu_2, \mu_3) \in \mathbb{C}^4 \setminus \{0\}$  such that

$$h(s, t) := (\lambda_2 s + \mu_2 t) \frac{\partial}{\partial x_2} f_c(s, t, 0, 0) + (\lambda_3 s + \mu_3 t) \frac{\partial}{\partial x_3} f_c(s, t, 0, 0) = 0.$$

So up to sign, we have two equal Polynomials. Since  $\frac{\partial}{\partial x_2} f_c(s, t, 0, 0)$  is a homogeneous polynomial of degree 2 in 2 variables, it can be factorized into 2 linear factors and one of them must also be a factor of  $\frac{\partial}{\partial x_3} f_c(s, t, 0, 0)$ . That means that there exists a point  $p = [p_0 : p_1 : 0 : 0]$  in  $L = (s, t, 0, 0)$  such that  $\frac{\partial}{\partial x_2} f_c(p) = \frac{\partial}{\partial x_3} f_c(p) = 0$ . However,  $L$  lies in our cubic and that means that  $f_c(s, t) = 0$  for all  $s, t$  meaning it never changes and therefore its derivative in  $s$  and  $t$  direction is constant zero. That leads to the contradiction, since in  $p$  all derivatives  $\frac{\partial}{\partial x_0} f_c(p) = \frac{\partial}{\partial x_1} f_c(p) = \frac{\partial}{\partial x_2} f_c(p) = \frac{\partial}{\partial x_3} f_c(p)$  vanish, giving us a singular point in  $p$ . Since that isn't possible, the Jacobian is invertible, and we are locally a graph of a continuous function.

So for a given  $c$ , that is for a given smooth cubic, and a line  $L$ , that Line can either be in  $\mathcal{M}$ , or not. If it is in  $\mathcal{M}$ , then locally around  $c$  all smooth cubics have 1 line close to  $L$ . If  $L$  is not in  $\mathcal{M}$ , then there is a neighbourhood of  $(c, L)$ , in which no Point lies in  $\mathcal{M}$ , meaning that no lines just come out of thin air. So locally the amount of lines doesn't change and, since  $U$  is a connected space, globally it doesn't change. Now the Fermat cubic has 27 lines and that's why very smooth cubic in  $\mathbb{P}^3$  has 27 lines.  $\square$

The used proof is rather beautiful by connecting algebra with the analytical category via the implicit function theorem. However, it is lacking in generality concerning the field. One could try to generalize it by using the implicit function theory for arbitrary

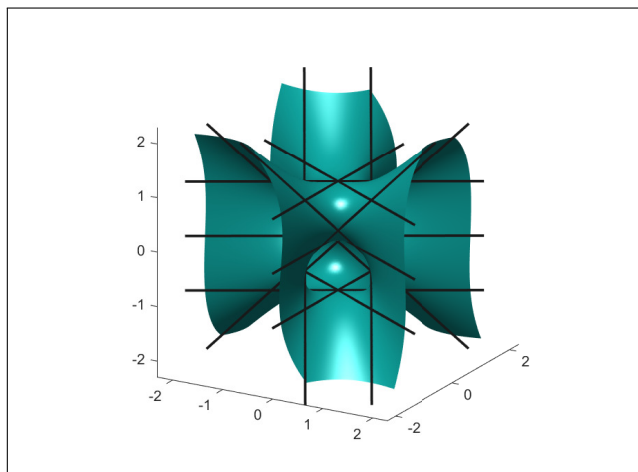


Figure 5: cubic surface with 15 lines

Banach spaces, opening the door to such fields that can be made into a complete metric space. But if one wants to follow that way, the difficulty arises in showing that the space of smooth cubics is connected concerning the topology used for the Banach space. Another generalization could be done for such polynomials that live in a field extension of  $\mathbb{C}$ , but are given by a polynomial with complex coefficients. The overview of such a proof would be as follows: The cubic  $X = V(f_c)$  interpreted over  $\mathbb{C}$  would have 27 lines, meaning that the variety of lines (or rather that scheme) in that cubic (as a variety in the Grassmanians), has 27 elements. Now those elements can be counted by the dimension of the first homology, which is invariant under field extensions, giving us that our variety  $X$  over the field extension would also have 27 lines. Or at least up to multiplicity. Now this idea was formulated, needing the language of schemes and other tools. This next small chapter will lift our understanding of algebraic variety's into the setting of schemes.

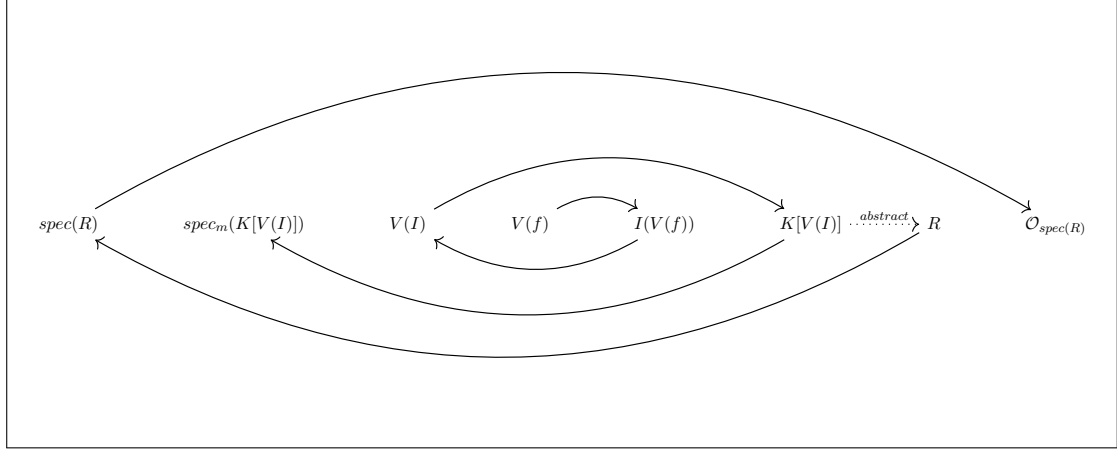


Figure 6: Abstraction of Algebraic geometry

## 4 Generalising results to different Fields

As already mentioned, we want to establish a more general notion of algebraic variety. The aim will be to establish the spaces which contain exactly 27 lines, and to give an introduction into schemes, that are the basics for further discussions of our mathematical field. We will consider this last goal archived once we have defined *schemes of finite type over an algebraically closed field*, that are used in the book [EH16], which can be seen as an overview into the theory of enumerative algebraic geometry. We will not prove the theorems needed to define those basics, but use examples and intuitive arguments to convince ourselves.

**Remark 7** (Abstracting varieties and defining sheafs). This construction can be viewed as an application of the hermeneutic circle translated to the world of mathematics, as shown in figure 6. We will look at a geometric object through algebraic lenses. Then in the world of pure algebra we will abstract it and see how we get back to the geometry, where we will again push it into the algebra to abstract it. Whilst abstracting, we want to be careful to fulfil the principle of permanence, such that we don't lose properties. In the end, we will connect the two world using the language of *ringed spaces*.

To start this path, let's think affine and in the end glue affine varieties to get back into the projective or even bigger world. So for a given variety  $X = V(J) \subset \mathbb{A}_K^n$ , where  $K$  is a field, and  $J$  is an ideal in  $K[x_1, \dots, x_n] = K[\underline{x}]$  we can look at the polynomial functions that are defined on  $X$ . Since we want to consider two functions equal if they coincide on  $V(J)$  we define the **coordinate ring of  $X$**  as follows:

$$K[X] := K[\underline{x}] / \sqrt{J}$$

Now all the information of  $X$  is hidden in  $K[X]$ , since we can get the variety back by taking the set of all maximal ideals called  $\text{spec}_m(K[X])$ . To get a more general

notion, we take the set of all prime ideals. We will call this the **spectrum of the ring  $\mathbf{K}[\mathbf{X}]$**  denoted by  $\text{spec}(\mathbf{K}[\mathbf{X}])$ . For an arbitrary ring  $R$  this spectrum will be the **affine scheme associated to  $R$** , and if  $R = \mathbf{K}[\mathbf{X}]$  it is the affine scheme associated to  $X$ . by taking prime ideals, we also consider subvarieties as points. This idea will be explained in detail in the remark 11. To go back to the world of algebra again, we need to look at functions defined on  $\text{spec}(R)$ . To do that, we denote the quotient field  $K(p) = \text{quot}(K/p)$  for  $p \in \text{spec}(R)$ . And for a  $f \in R$  we define  $f(p)$  as the image under the mapping  $R \rightarrow K/p \rightarrow K(p)$ . As a result, we get that  $f(p) = 0$  iff  $f \in p$ . Using this notion, we can extend the Zarisky topology and vanishing ideals into the new world, doing the following. For a given ring  $R$  and  $S \subset R$  we define

$$\begin{aligned} V(S) &:= \{p \in \text{spec}(R) \mid f(p) = 0 \ \forall f \in S\} \subset \text{spec}(R), \\ I(X) &:= \{f \in R \mid f(P) = 0 \ \forall P \in X\}. \end{aligned}$$

With those definitions all the inclusion reversing properties are preserved, so we can get the Zarisky topology defined on our affine scheme. Our now constructed topological space  $\text{spec}(R)$  can be taken back into the algebraic world in a way more detailed way. Instead of looking at its global coordinate ring, which is  $R$ , we can ask ourselves, what are the functions defined on open subsets of  $\text{spec}(R)$ . To do that, we need the notion of regular functions as locally rational functions. A regular function  $\phi_U$  on an open subset  $U \subset \text{spec}(R)$  will be a family of  $(\phi_p)_{p \in U}$ , where the  $\phi_p$  are locally quotients of elements in  $R$ , without dividing by 0. So for all  $p \in U$  there exists  $g, f \in R$  such that there is an open neighbourhood  $U_p$  of  $p$  such that for all  $Q$  in  $U_p$ ,  $f(Q) \neq 0$  and

$$\phi_Q = \frac{g}{f}.$$

For a given  $U \subset \text{spec}(R)$  we can now look at the  $\mathbf{K}$ -algebra of regular Functions on  $U$  called  $\mathcal{O}_{\text{spec}(R)}(U)$ . In fact, this  $\mathcal{O}$  will be a functor into the category of rings, called the **structure sheaf on  $\text{spec}(R)$**  fulfilling the following properties.

1. For an inclusion  $V \subset W$  in the topological space, there exists a restriction morphism  $\rho_V^W : \mathcal{O}_W \rightarrow \mathcal{O}_V$  such that for  $U \subset V \subset W$ :  $\rho_U^W = \rho_U^V \circ \rho_V^W$ .
2.  $\mathcal{O}(\emptyset) = 0$
3. For any open covering  $U = \bigcup_{i \in I} U_i$  we have :
  - a) for  $f$  in  $\mathcal{O}(U)$  with  $\rho_{U_i}^U(f) = 0$  for all  $i$ , then  $f = 0$  in  $\mathcal{O}(U)$
  - b) for any collection of  $f_i$  in  $\mathcal{O}(U_i)$ , such that  $\rho_{U_i \cap U_j}^{U_i}(f_i) = \rho_{U_j \cap U_i}^{U_j}(f_j)$  for all  $(i, j) \in I^2$  there exists a unique  $f \in \mathcal{O}(U)$  such that  $\rho_{U_i}^U(f) = f_i$ .

In the following, we will denote the restriction of an element  $\phi$  of the ring to a subset  $V$  with  $\phi|_V$ .

So now this functor is our new translation into the algebraic world. To understand this further, we can look at a point  $P$  in  $\text{spec}(R) = X$ . If we want to understand the

functions locally around  $P$  we can look at all elements in  $\mathcal{O}(U)$ , where  $U$  is an open neighbourhood of  $P$ . To not get any locally similar functions, we consider 2 functions equivalent, if there is a shared open neighbourhood  $V$  of  $P$ , where the two functions are equal. So taking the quotient of  $\bigcup_{U \ni P} \mathcal{O}(U)$  defined by that relation gives us the desired space. We will call that the **stalk**  $\mathcal{O}_{X,P}$  and following the agricultural naming we will call its elements the germs. It can be shown that the stalk  $\mathcal{O}_{\text{spec}(R),P}$  in  $P$  is isomorphic to  $R_P$ , where  $R_P$  is the localization of  $R$  in the prime ideal  $P$ . So each stalk is a local ring. Furthermore, one can show that for the open set  $D(g) := X \setminus V(f)$ ,  $\mathcal{O}_{\text{spec}(R)}(D(g)) = R_g$ . The latter shows that we get our ring or in the classical sense coordinate ring back by looking at  $g = 1$ . To bring both worlds together, we define **locally ringed spaces** as tuples of topological spaces and sheaves, such that every stalk is a local ring. The above shows that we can view affine schemes as locally ringed spaces. From now on we will abuse the notations and  $\text{spec}(R)$  will also denote the ringed space  $(\text{spec}(R), \mathcal{O}_{\text{spec}(R)})$ . So far we have generalized varieties to be sets of prime ideals, and we have expanded our translation to the algebraic world. To really expand the categories of varieties, we need to view new elements as varieties. To do that, we will look at arbitrary locally ringed spaces that are isomorphic to an affine scheme. So our next step is to understand morphisms of locally ringed spaces.

**Remark 8** (multiplicity of intersections). However, let's first look at what properties we gained or lost by abstracting so far. Since this thesis looks at numbers of intersections of tangents, we will explore our new world by looking at the intersection of  $V(x^2 - y)$  and  $V(y)$  in  $\mathbb{C}^2$ . We know that in the classical definition we have an intersection  $P = V(\langle x^2 - y, y \rangle) = V(x^2) = V(x)$ . With that view we don't know that our intersection is a tangent or in other words if we move  $V(y)$  infinitesimal we would get 2 intersections. So let's calculate the intersections in our new definitions. We know that

$$\text{spec}(K[x, y]/x^2 - y) \cap \text{spec}(K[x, y]/y) = \text{spec}(K[x, y]/\langle x^2 - y, y \rangle) = \text{spec}(K[x]/x^2).$$

So the intersection is the tuple  $(\text{spec}(K[x]/x^2), \mathcal{O}_{\text{spec}(K[x]/x^2)})$  and our topological space still contains only 1 element, since there is only 1 prime ideal in  $K[x]/x^2$ .

That is because for any prime ideal  $p$   $x \cdot x = 0 \in p$  and therefore  $x \in p$  and since  $K[x]/x$  is a field,  $\langle x \rangle$  was already maximal.

Let us now define  $\mathcal{R} := K[x]/x^2$  and look at the dimension of  $\mathcal{O}_{\text{spec}(\mathcal{R})}(R) = K[x]/x^2 = \mathcal{O}_{K, (0,0)}(K)/x^2$  as a  $K$  vector space, we get 2. Motivated by that calculation, we will define the **intersection multiplicity** of  $n$  hypersurfaces  $Z_i = V(f_i)$  in a smooth  $n$ -dimensional variety  $X$ , that intersect in a 0-dimensional variety as follows:

$$\text{mult}_x(Z_1 \cap Z_2) = \dim_k(\mathcal{O}_{X,x}/\langle f_1, \dots, f_n \rangle).$$

This definition will work for schemes in a scheme  $X$  over  $k$ . But let's get concrete and look at 3 varieties  $V_i = V(f_i)$  in  $\mathbb{P}^3$ . One can show that  $\mathcal{O}_{\mathbb{P}^3, p} \cong C[x_0, x_1, x_2]_{\mathfrak{m}_p}$ , where  $p \in V(x_0)$ . Using that, we can see that our notion of multiplicity is the dimension of the ring  $K[x_0, x_1, x_2]_{\mathfrak{m}_p}/\langle f_1(1, x_1, x_2), f_2(1, x_1, x_2), f_3(1, x_1, x_2) \rangle$ . That is the space of



rational functions on the intersection, where the localization makes sure we are well-defined in  $p$ . Not taking the radical increases the dimension of the space, which in a sense is the core of the multiplicity.

**Remark 9.** Since  $\mathcal{O}_{\text{spec}(\mathcal{R})}(\text{spec}(\mathcal{R})) = \mathcal{R} = K[x]/x^2$ , we know that the global regular functions have the form  $f = a + bx$  for  $a, b$  in  $K$ . However, since they only take on a value for  $f(\langle x \rangle)$ , the regular functions in  $\mathcal{O}(U)$  for an open subset  $U$ , are not just set theoretical functions. That shows us, that if we want to define morphisms on locally ringed spaces, we need to tell what to do with a regular functions in a way that does not just look at them as functions. We need to consider and define them as elements in the  $K$ -algebra.

**Definition 7** (Schemes). To understand schemes as locally isomorphic, we have to understand **isomorphisms of locally ringed spaces**. A **morphism of ringed spaces**  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  needs to contain the following data:

1. A continuous function  $f : X \rightarrow Y$ .
2. For every open subset  $U \subset Y$  a ring morphism  $f^* : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U))$  called the **pullback on  $U$** , such that:

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{f_V^*} & \mathcal{O}_X(f^{-1}(V)) \\ \text{a) The pullback commutes with restrictions.} & \downarrow \rho_U^V & \downarrow \rho_{f^{-1}(U)}^{f^{-1}(V)} \\ \mathcal{O}_Y(U) & \xrightarrow{f_U^*} & \mathcal{O}_X(f^{-1}(U)) \end{array}$$

This requirement ensures well-defined maps between stalks.

- b) Finally, we require for every stalk that the induced map  $f_P^* : \mathcal{O}_{Y,f(P)} \rightarrow \mathcal{O}_{X,P}$  is a local homomorphism, meaning that the preimage of the maximal ideal in  $\mathcal{O}_{X,P}$  is the maximal ideal in  $\mathcal{O}_{Y,f(P)}$ .

With our now constructed definition of morphism, we can define schemes as the following: A **scheme**  $X$  is a (locally) ringed space with an open cover  $X = \bigcup_i U_i$  such that each  $U_i$  is isomorphic to an affine schemes. The notion of open cover works, since the sheaf has to work with restrictions.

**Remark 10.** By the last definition, we got rid of the restriction of being affine. If we look at the definition of projective variety so far, we viewed them as zero-sets of homogeneous polynomials  $f_1, \dots, f_r$  in  $K[x_0, \dots, x_n]$ . Now we can look at them as subsets of  $(K \setminus \{0\})/K$  with the topology given by zero-sets of homogeneous polynomials. For such a space we can get the open affine neighbourhood of  $p$ , where the  $i$ th coordinate of  $p$  is non-zero, by cutting out  $V(x_i)$ . We then get an affine open set, with the structure sheaf over

$$R = K[x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n] / \langle f_j(x_0, \dots, x_{i-1}, 1, \dots, x_n) \mid 0 \leq j \leq n \rangle.$$

In example 4, we will get to know a class of schemes, that can be seen as a generalization of projective space.

**Definition 8.** For a given scheme  $Y$  we can look at the **scheme over  $Y$** . That is just a scheme  $X$  equipped with a morphism to  $Y$ . If  $Y = \text{spec}(R)$  for a ring  $R$ , we will call it a scheme over  $R$ .

**Example 3.** When we used to look at a variety,  $V(x^2+1)$  we had to clarify which Field is the underlying one. As a scheme, we can now look at  $\text{spec}(\mathbb{R}/\langle x^2+1 \rangle) = \text{spec}(\mathbb{C}) = \{\langle 0 \rangle\}$  as a topological space. Or we look at  $\text{spec}(\mathbb{C}/\langle x^2+1 \rangle) = \{\langle x+i \rangle, \langle x-i \rangle, \langle 0 \rangle\}$ . So we got rid of an embedding and look at our varieties as independent spaces.

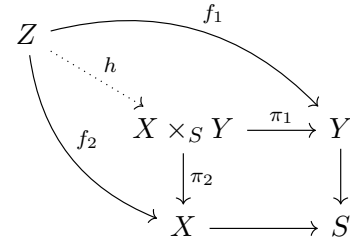
We can get the affine space  $\mathbb{A}_K^n$  over  $K$  by looking at  $\text{spec}(K[x_1, \dots, x_n])$ . However, this space will consist of classical closed points given by maximal ideals, a point  $\langle 0 \rangle$  called the generic point whose closure is the whole space. That is since  $V(I(\langle 0 \rangle)) = V(\{f \in K[x_1, \dots, x_n] | f(\langle 0 \rangle) = 0\}) = V(\langle 0 \rangle) = \text{spec}(K[x_1, \dots, x_n])$ . And lastly, all subvarieties consisting of points and their subvarieties will be points in  $\mathbb{A}_K^n$ .

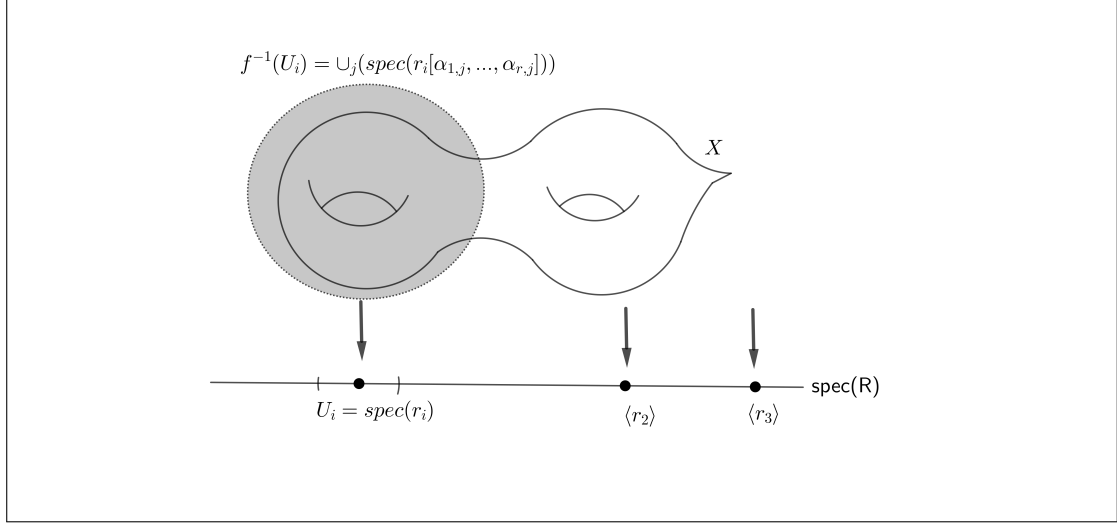
Lastly, to understand schemes over a field  $k$ , we just have to look at the pullback. If we have a scheme  $X$  over  $k$  we have a continuous map  $\phi : X \rightarrow \text{spec}(k) = \{\langle 0 \rangle\}$ . since everything maps to  $\langle 0 \rangle$ , we have only one pullback, that is a ring homomorphism  $\phi^* : k \rightarrow \mathcal{O}_X(X)$ . This lets us view the sheaf as one mapping into a  $k$ -Algebra. This idea works also for rings, where we can look at preimages and therefore a scheme over another scheme can be viewed as a family of schemes over certain rings. This idea can be seen in figure 7.

**Remark 11** (understanding schemes over  $k$ ). In example 3 we looked into the scheme  $\mathbb{A}_K^n$  and realized that it consisted of all subvarieties of the classical affine space  $K^n$ . This construction will give us a fully faithful functor  $\tau : \mathfrak{Var}(k) \rightarrow \mathfrak{Sch}(k)$  from the category of affine varieties over  $k$  into the category of affine schemes over  $k$ , that can be extended to the category of varieties, to the category of schemes. That gives us an interpretation of schemes as varieties. A variety in this context will be a quasi-projective one, meaning a projective variety in  $\mathbb{P}^n$  or an open subset. The extending of the functor works since varieties and schemes are covered by affine ones.  $\tau(X)$  maps a topological space to the set of all closed subsets. On  $\tau(X)$  we can define a topology by taking as closed sets the subsets of the form  $\tau(Y)$ , where  $Y$  is a closed set in  $X$ . Then  $\tau$  is a functor on topological spaces. We can now define a continuous map  $\alpha : X \rightarrow \tau(X)$ , sending  $P$  to  $\alpha(P) := \{P\}$ . One can now show that  $(\tau(X), \alpha^* \mathcal{O}_X)$  is a affine scheme over  $k$ . The proof can be seen in [Har97, p. 78 II 2.6]. In fact, the objects in the image of that functor are integral and separated schemes over  $k$  of finite type. Integral means that for all  $U \subset X$   $\mathcal{O}_X(U)$  is a integral domain, and separated is a notion inspired by the topological property of being Hausdorff.

**Definition 9** (Separated scheme).

To define separated, we first define the fibre product of 2 schemes  $X, Y$  over a scheme  $S$  to be a scheme  $X \times_S Y$ , together with morphisms  $\pi_1 : X \times_S Y \rightarrow X$  and  $\pi_2 : X \times_S Y \rightarrow Y$ , that makes the following diagram commute. The existence of such a product is shown in [Har97, p. 87 II 3.3]




 Figure 7: Scheme over  $R$  of finite type

For a given scheme  $X$  over,  $S$  we can now look at the unique morphism  $\Delta : X \rightarrow X \times_S X$  coming from the universal property, such that  $\pi_i \circ \delta = id_X$ . We call  $X$  separated over  $S$  if  $\Delta$  is a closed immersion. This definition is inspired from the definition of a topological space  $X$  being hausdorff, iff the diagonal is a closed subset of  $X \times X$ .

**Definition 10** (Finite type). A morphism of schemes  $f : X \rightarrow Y$  is of finite type, if there is a affine open covering  $U_i \subset Y$  of  $Y$  where the  $U_i = \text{spec}(B_i)$ , such that  $f^{-1}(U_i)$  can be covered by finitely many affine open subsets  $V_{i,j}$ , where the  $V_{i,j} = \text{spec}(A_{i,j})$ , such that the  $A_{i,j}$  are finitely generated  $B_i$ -algebras. This construction can be seen in figure 7.

**Remark 12** (Smoothness). Lastly, we can translate the notion of being smooth into the world of schemes. So far, we have defined smooth only for classical projective hypersurfaces, by requiring that the Jacobian only vanishes at 0. For an (affine) variety  $X$  in general, we would require the tangent space at every point has the dimension of  $X$ . Since every variety can be covered by an affine one, or is isomorphic to one, we will translate that definition into the world of schemes. To do that translation, we will consider an affine variety, given by  $f_1, \dots, f_r$  going through the origin. The tangent space at the origin of a variety is given by all the linear parts of the  $f_i$ , respectively as elements in  $K[X]$ . For arbitrary points, we can make a transformation and use the linear terms of the formal taylor expansion. Those classes of linear polynomials  $[f_i]$  we can map into  $\mathfrak{m}_0/\mathfrak{m}_0^2$ , where  $\mathfrak{m}_p$  denotes the maximal ideal corresponding to  $p$  in  $\mathcal{O}_p \cong \mathcal{O}(X)_{\mathfrak{m}_p}$ . This mapping takes the  $f_i$  and interprets them first as  $\frac{[f_i]}{1}$  in  $\mathfrak{m}_p$ . That is possible, since they are linear and therefore vanish at 0, and 1 isn't in  $\mathfrak{m}_0$ . Finally, we can just take the classes of those and get  $\left[\frac{[f_i]}{1}\right] \in \mathfrak{m}_0/\mathfrak{m}_0^2$ . One can show that this mapping is an isomorphism, and

this construction works for arbitrary  $p$ . This isomorphism is an isomorphism of vector spaces, where  $\mathfrak{m}_p/\mathfrak{m}_p^2$  is a vector space over the residue field  $\mathcal{O}_p(X)/\mathfrak{m}_p$ . The well-definition of the multiplication is clear, since  $(a + \mathfrak{m}_p) \cdot (b + \mathfrak{m}_p) = a \cdot b + a\mathfrak{m}_p + b\mathfrak{m}_p + \mathfrak{m}_p^2 = a \cdot b$ . So a classical variety  $X$  of dimension  $n$  is smooth, iff  $\dim_{\mathcal{O}_p(X)/\mathfrak{m}_p}(\mathfrak{m}_p/\mathfrak{m}_p^2) = n$  for all  $p$ . This is exactly the property we will use to define regular schemes:

**Definition 11** (Smooth schemes). **A scheme  $X$  is regular or smooth**, iff all of its local rings  $\mathcal{O}_p$  are regular, meaning that

$$\dim_{\mathcal{O}_p(X)/\mathfrak{m}_p}(\mathfrak{m}_p/\mathfrak{m}_p^2) = n.$$

**Example 4** (Projective schemes). So far we understood affine schemes over  $k$  and schemes that are glued from affine ones or are locally affine ones. But what exactly are projective varieties? In the classical world. We took the graded ring  $K[x_0, \dots, x_n]$  and looked at zero sets of homogeneous polynomials. However, let's go the route of coordinate rings again. If we want the projective space, we cannot simply take the  $\text{spec}(S)$ , where  $S$  is a graded ring. That would give us the affine space. However, we can take  $\text{Proj}(S) := \{\mathfrak{p} \subset S \mid \mathfrak{p} \text{ is a homogeneous ideal in } S\}$ . For a homogeneous ideal  $\mathfrak{a}$ , we can define the subset  $V(\mathfrak{a}) := \{\mathfrak{p} \in \text{Proj}(S) \mid \mathfrak{p} \subset \mathfrak{a}\}$ . We can view that definition as analogous to the definition of  $V(S)$  in Definition 7, without the use of a function. One can show that those  $\bigcap_i V(\mathfrak{a}_i) = V(\sum \mathfrak{a}_i)$  for families  $\{\mathfrak{a}_i\}$ , and  $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$ , and with that, we can define a topology on  $\text{Proj}(S)$ . With that topology we can define a sheaf as usually by sending open subsets  $U \subset \text{Proj}(S)$  to  $\mathcal{O}(U)$ , where  $\mathcal{O}(U)$  is the ring of function that are of the following form: A function  $\phi_U \in \mathcal{O}(U)$  is a family  $(\phi_P)_{P \in U}$ , where the  $\phi_P$  are elements of the localized ring  $S_{\mathfrak{p}}$ , such that they are local quotients of elements of  $S$ : So for every  $\mathfrak{p} \in U$  there is a open neighbourhood  $V \subset U$  of  $\mathfrak{p}$  and homogeneous elements  $g, f$  of  $S$ , of the same degree, such that for all elements  $\mathfrak{q} \in V$   $f$  is no element in  $\mathfrak{q}$  and  $\phi_{\mathfrak{q}} = \frac{g}{f}$ . This construction is closely related to the structure sheaf of affine varieties, with the restriction of the local quotient being of the same degree. With the usual restrictions and the local construction, it is clear, that this is a sheaf. To see that it is a scheme, we need to show that it has an open covering such that the open subsets are affine schemes. The open subsets are also as one might expect. If  $S_+ = \bigoplus_{d>0} S_d$  is the ideal without the 0 degree homogeneous polynomials, we can look at the open subsets  $D(f) = \{\mathfrak{p} \in \text{Proj}(S) \mid f \notin \mathfrak{p}\}$  for any  $f \in S_+$ . Now the open sets  $(D(f), \mathcal{O}_{\text{Proj}(S)}|_{D(f)})$  form a covering and are isomorphic to  $\text{spec}(S_f)$ . This is exactly what one would expect, and the construction in example 10 reflect that by arguing for the  $D(x_i)$ . We will now call a projective subscheme of  $\mathbb{P}^n$  a scheme that is of the form  $\text{Proj}(K[x_0, \dots, x_n]/I)$  for a homogeneous ideal  $I$  of positive degree. A proof can be found in [Har97, p. 76 II 2.5].

**Definition 12** (Dimension). We define the dimension of a scheme to be the topological dimension of  $X$ . That is, as the maximal length of a chain of included closed subspaces.

**Remark 13.** For a projective subscheme  $X = \text{Proj}(S)$  of  $\mathbb{P}^n$ , the subspaces are given as a collection of ideals contained in a homogeneous ideal  $\mathfrak{a} \subset S$ . So the dimension of  $X$  can

be calculated as the maximal length of an ascending chain of ideals. By Krull's principal ideal theorem, we know, that the height or codimension of a prime ideal  $I$  is at most 1. That means by definition that the longest chain of prime ideals contained in  $I$  is 1. We can extend the chain to a maximal chain of prime ideals  $P_0 \subset I = Q_0 \subset \dots \subset Q_r$  of  $S$ , where all ideals  $Q_i$  contain  $I$ . Therefore, the longest chain of ideals in  $R/I$  is  $r = \dim(S) - 1$ . So a cubic in  $\mathbb{P}^3$  is nothing but  $\text{Proj}(K[x_0, \dots, x_3]/\langle f \rangle)$ , where  $f$  is homogeneous of degree 3. If  $f$  is homogeneous of degree 1 we get a line in  $\mathbb{P}^3$ . Note that this notion of degree only works for hypersurfaces. In general, one can define the degree by usage of a Hilbert polynomial. However, in this thesis we mostly care for hypersurfaces and the general definition coincides on those with our simple one, as seen in [Har97, p. 52 I.7].

**Theorem 3** (Bézout's theorem). *Let  $X$  be a projective subscheme of  $\mathbb{P}^2$  of positive dimension, and let  $f \in K[x_0, \dots, x_n]$  be a homogeneous polynomial such that no topological irreducible subset of  $X$  is contained in  $Z(f) := \text{Proj}(K[x_0, \dots, x_n]/\langle f \rangle)$ . Then*

$$\text{length}(X \cap Z(f)) = \sum_{z \in X \cap Z(f)} \text{mult}_z(X \cap Z(f)) = \deg(X) \cdot \deg(Y),$$

where the length counts the points of  $X \cap Z(f)$ , with their scheme-theoretic multiplicities.

*Proof.* A proof can be found in [Gat02, p. 95 6.2]. We just need to translate the notion of degree of a 0-dim scheme into the notion of length.  $\square$

**Theorem 4** (There are exactly 27 lines on smooth cubics). *Every projective subscheme of  $\mathbb{P}^3$  over  $k$ , given by an ideal of degree 3 over an algebraically closed field of characteristic 0, contains exactly 27 distinct lines.*

*Proof.* A proof can be read in [EH16, p. 166 5.1] or in [Har97, p. 402 4.9]. Both use completely different approaches. Whilst Hartshorne relies heavily on the theory of divisors, Harrison shows that the degree of the fourth Chern class of the space of cubic forms on a line is 27.  $\square$

With those definitions, we have established the basic language of algebraic geometry. For the next chapter, we will go back to the world of varieties as zero-sets of polynomials. However, we now know, that we can view them as schemes and some concepts like the multiplicity of intersections, Bézout's theorem or morphisms of schemes will still be needed.

## 5 The 28 bitangents

To study lines connected to algebraic surfaces, we introduced the Grassmannian. In this chapter, we want to get to know the lines touching a variety. To study that, we will first introduce the polar of a variety from a point and latter we will get to know the dual curve, giving an algebraic way to study tangent lines. The goal of this chapter is to get a different view on lines and to see why there are always 28 bitangents on a (smooth) quartic curve in  $\mathbb{P}^2$ . For a given radical ideal defined by  $f \in K[x]$  we define the tangent in a point  $p$  as the variety given by  $\sum_i x_i \cdot \frac{\partial f}{\partial x_i}(p)$ . So a point  $q$  lies in the tangent of a hypersurface  $V$  iff  $\sum_{i=0}^2 q_i \cdot \frac{\partial f}{\partial x_i}(p) = 0$ . So if we want to ask if there is a tangent going through a certain point outside the variety, we need to turn the equation around. Instead of calculating the tangent, we search for points  $p$  in the variety, where the tangent goes through  $q$ . That however is pretty easy since we can just go through all points  $p$  in our  $V$  and check if  $p$  is in  $V\left(\sum_{i=0}^2 q_i \cdot \frac{\partial f}{\partial x_i}\right)$ . For this chapter, we will work over the complex field again and follow ideas from [Fis94].

**Definition 13** (The polar). For a Hypersurface  $X = V(f)$  in  $\mathbb{P}^n$  we define the polar  $P_q X$  of a point  $q$  in  $\mathbb{P}^n$  as the variety given by

$$\Delta_q f := \sum_{i=0}^2 q_i \cdot \frac{\partial f}{\partial x_i}.$$

Figure 8 shows the polar of the Trott curve defined by  $f = 144(x^4 + y^4) - 225z^2(x^2 + y^2) + 350x^2y^2 + 81z^4 = 0$  in the chart  $z = 1$ , in  $q = [0 : 0 : 1]$ . Using this polar, we

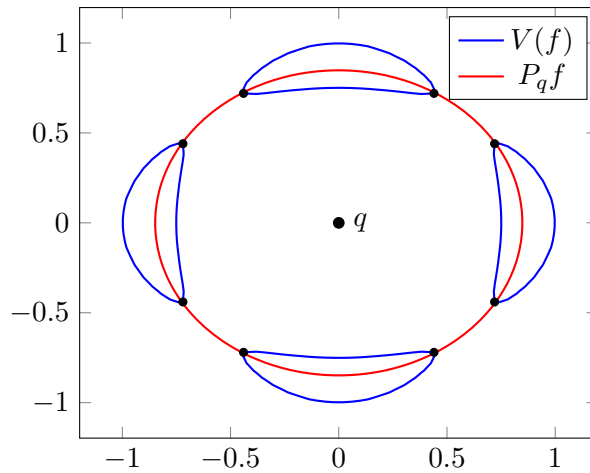


Figure 8: The polar of a quartik

can move on to create a space, where the elements correspond to lines. To get there, we will change the roles of points and lines. Whilst in classical (axiomatic) plane projective geometry there are points, lines that uniquely defined by two points and a relation

explaining when a point is in a line, we create a space of lines and points, and its converse relation. We will call that the dual projective space. Whilst this construction can be generalized to arbitrary dimension, we will only need the dual plane. To coordinate that construction, we will use the polar and the most simple nonlinear curve  $V(x_0^2 + x_1^2 + x_2^2)$ . The polar is obviously given by a polynomial of degree  $n - 1$ . Note that this polynomial might not be minimal. In terms of Fischer, its degree is the degree of the effective divisor. Since we established Bézout's theorem for schemes, we don't need to worry about that, since we don't need our curves to be defined by radical ideals.

**Definition 14.** (Dual plane) A point  $y = [y_0 : y_1 : y_2]$  in the dual plane  $(\mathbb{P}^2)^*$  is the polar of  $y$  on the curve  $V(x_0^2 + x_1^2 + x_2^2)$  in  $\mathbb{P}^2$ . By calculating that polar, we realize that  $y$  is identified with the line  $y_0x_0 + y_1x_1 + y_2x_2 = 0$ . So we already see, that our new geometric space consists of lines, and our coordinates are uniquely defined up to multiples.

To deepen our understanding of it from the viewpoint of classical geometry, we can also consider points, as the intersection of two lines. This construction leaves points uniquely defined by two lines and a line is in relation with the point, if the point is on the line. So we use the converse relation and therefore end up with another classical projective space, in which the points represent lines. The next definition will make clear, why this construction is useful to study tangents and bitangents.

**Definition 15.** For a given plane curve  $V$  in  $\mathbb{P}^2$ , we define the dual curve

$$V^* := \{y \in (\mathbb{P}^2)^* \mid y \text{ is a tangent of } V\}.$$

Note that this construction tells us that we get points in  $(\mathbb{P}^2)^*$  from lines in  $\mathbb{P}^2$ . If we want to understand the relation between the plane and its dual, we can further inspect the mapping. So far we know, that a line  $l$  in the plane will be sent to the point in the dual, that reflects its gradient  $(q_1, q_2, q_3)$ . The  $q_i$  are the  $i$ -th partial derivative, and it is clear, that we get the line back when we go back to the plane by constructing  $q_1x_1 + q_2x_2 + q_3x_3 = 0$ .

We can also think about  $q$  as being the orthogonal Vector of the origin plane  $l$  in  $K^3$ . If we take all lines in  $\mathbb{P}^2$  going through a point, that is all planes in  $K^3$ , that include the line defined by  $q$ , and map them to their orthogonal origin lines, we get a new plane in  $K^3$  respectively a line in  $(\mathbb{P}^2)^*$ , where each point comes from a line in  $\mathbb{P}^2$  going through  $q$ . These mappings can be seen in figure 9.

One can show, that the dual curve  $V^*$  is an algebraic curve of degree  $n^*$ , where  $n^*$  denotes the maximal amount of tangents on  $V$  through one point. To show the latter, we can view a point  $q$  in  $\mathbb{P}^2$  as the image of a line  $q^*$ , that came from the bundle of all lines through  $q$  in  $\mathbb{P}^2$ . The intersection of  $V^* \cap q^*$  consists of all those points that coincide with tangents going through  $q$ . Bézout's theorem now tells us, that our dual curve has a degree  $n^*$  of the amount of tangents and their multiplicity. We also know, that most points  $q$  have exactly  $n^*$  tangents, since we get a higher multiplicity in our intersection  $V^* \cap q^*$  iff  $q^*$  is a line connecting any point  $q^*$  with the intersection of the

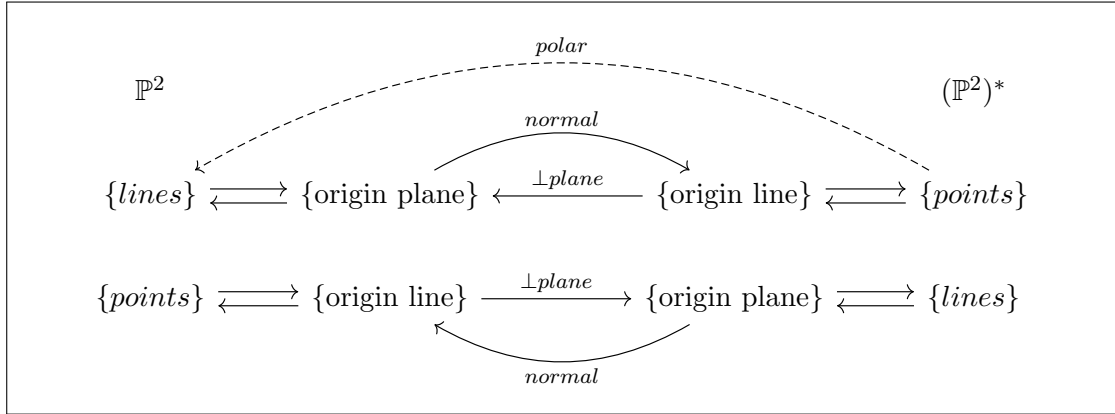


Figure 9: The dual Plane

polar of  $D_{q^*}(I(V^*))$ . Since the polar and its variety have only finite intersection, that is almost never the case. This just reformulates, that for a randomly chosen point there is almost never a double tangent or tangent through an inflection point.

**Remark 14.** As one might notice, we encounter the notion of multiplicity of intersections. In remark 8 we defined the intersection multiplicity in  $x$  of two subschemes of codimension 1 given by  $f_1, f_2$  in another scheme  $X$  as the  $K$ -dimension of  $\mathcal{O}_{X,x}/\langle f_1, f_2 \rangle$ . Fischer, whom we follow in this section, only needs to define the multiplicities for intersections in  $\mathbb{P}_{\mathbb{C}}^2$  and defines it as the multiplicity of a root of the resultant  $R_{f_1, f_2}$ . Our definition can be seen as a generalization of the ladder and coincides with it. That is shown in [BCG09] Corollary 4.3.9. Note that the resultant is not invariant for the chosen polynomial that defines a certain curve, same as the intersection multiplicity changes if we have a scheme given by a non-radical ideal. We will later intersect with polars or Hessians, that are given by non-minimal polynomials. So we don't want to get rid of that and view the multiplicity more as an algebraic construction, than a geometrical one.

If we want to calculate the multiplicity of a line  $L = V(x_2)$  with a curve  $C = V(f)$ , we can restrict  $f$  to  $L$ , and calculate the multiplicity of its root in the intersection. In practice, we can sort  $f$  with respect to  $x_2$ , such that

$$f = f_0 x_2^n + f_1 x_2^{n-1}, \dots, f_n,$$

and the  $f_i$  are homogeneous. The intersections of  $C$  and  $L$  are the roots of  $f(x_0, x_1, 0) = f_n$ . If  $f_n = 0$ , we know that  $L \subset C$  and otherwise, the degree of  $f_n$  is  $n$ . We can now calculate the linear factor decomposition of  $f_n = \prod_i (a_i x_0 - b_i x_1)^{\alpha_i}$ , such that the  $\alpha_i$  sum to  $n$ . The multiplicity of an  $[a_i : b_i : 0]$  of the intersection  $C \cap L$  is the  $\alpha_i$  or the multiplicity of the root. One can show that this calculation is invariant under changes of coordinates, and therefore we can calculate the intersection multiplicity of a line and a curve like that. The multiplicity also behaves like one expect, meaning that 2 curves intersect with multiplicity 1 iff they have two different tangents [Gat21b, p. 16 2.19].



**Definition 16** (Order of a point and tangents). Before we can formulate the Plücker formulas, we will get to know singularities of curves. For a given point  $p \in C \subset \mathbb{P}^2$ , where  $C = V(f)$  is an algebraic curve and  $\langle f \rangle$  is radical, we define the order at  $p$  as follows. First do the multivariate formal Taylor series of  $f$  in  $p$ , after dehomogenising to the affine world:

$$f(x_1, x_2) = \sum_k f_{(k)}, \text{ where } f_{(k)} = \sum_{\mu+\nu=k} \left( \frac{1}{\mu!\nu!} \frac{\partial^k}{\partial x_1^\mu \partial x_2^\nu}(p) \right) (x_1 - p_1)^\mu (x_2 - p_2)^\nu$$

Then we get the order  $\text{ord}_p(C) = \min\{k | f_{(k)} \neq 0\}$ . Firstly, we can see, that the order is always greater than zero, since  $f_{(0)} = f(p) = 0$ . If we now inspect figure 10, we know, that  $V(f)$  is singular, and we cannot construct a classical tangent at its origin, whilst it is clear, that there would be two reasonable tangents. To get those, we simply define a line  $L$  to be a tangent in  $p$  if  $\text{ord}_p(C) < \text{mult}_p(C \cap L)$ . Intuitively, this definition is clear, as a tangent adds a layer to the point. If the order is 1, it is clear that we have a well-defined tangent, giving us a smooth point. So a point is singular, iff the order is greater than 1.

**Definition 17** (Characterization of singularities). We saw, that a point  $p$  in  $C$  is singular iff  $\text{ord}_p(C) \geq 2$ . Let  $f$  be a polynomial describing  $C$  around  $p = [1 : 0 : 0]$ . Minimal means, that  $\langle f \rangle$  is radical. We then have a sum of homogeneous polynomials  $f_k \in \mathbb{C}[x_1, x_2]$ :

$$f = f_{(2)} + f_{(3)} + \dots + f_{(k)},$$

where  $k$  is the degree of  $f$ .

For  $f_{(2)} = c_1x_1^2 + c_2x_1x_2 + c_3x_2^2$  if  $c_2^2 \neq 4c_1c_3$ , then  $f_{(2)}$  isn't just a square, and we therefore get 2 different linear factors, giving us 2 different tangents  $T$  in  $p$  as seen in figure 10. To see that computationally, can now transform our coordinates such that  $f_{(2)} = x_1x_2$ , by just using the linear factors as substitutions. It is now clear, that  $\text{mult}_p(C \cap T) \geq 3$ , since for  $V(x_1)$  we have the parametrization  $(t_1x_0, 0, t_2x_2)$ . Inserting that into  $f$ , we get  $f_{(2)} = 0$ , and therefore  $V(x_1)$  is a tangent. Analogously, we can show that  $V(x_2)$  is a tangent. If the multiplicity of the intersection with those tangents is exactly 3, we call  $p$  a *simple double point*. Fishher later shows that the intuition is true and this means, that there is no inflection point in the Intersection of the tangents. The two branches that meet in the point are for themselves smooth and the graph of a holomorphic function. So figure 10 is actually sufficient for an intuition. After homogenizing, we know that our  $C$  is given by

$$F = x_0^{n-2}x_1x_2 + G(x_0, x_1, x_2),$$

where  $\text{ord}_p(G) \geq 3$ .

Is on the other hand  $c_2^2 = 4c_1c_3$ , then  $f_{(2)}$  is a square with a double root. After a transformation, we can assume  $f_{(2)} = x_2^2$ .  $T = V(x_2)$  gives us a tangent, which we call simple, iff  $\text{mult}_p(T \cap C) = 3$ . Again, Fischer later shows, that around  $p$  there is a parametrization of the form  $t \mapsto (t^2, t^3 + \alpha_4t^4 \dots)$ , giving us a cusp as seen in figure 11.

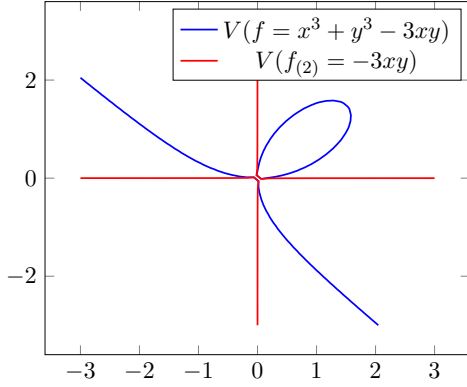


Figure 10: Folium of Descartes

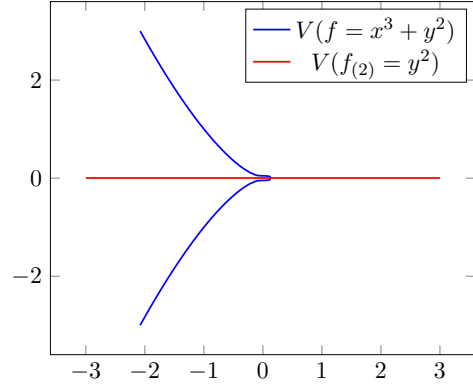


Figure 11: Tangent at Cups

**Remark 15.** If a curve  $C = V(f)$  has a simple tangent in  $p$ , and  $q \in T$  with  $q \neq p$ , then the polar  $T_q C$  intersects transversal, meaning that  $\text{mult}_p(T_q C \cap C) = 1$ .

*Proof.* A proof can be found in [Fis94, p. 46 4.3]. Intuitively, one shows, that the polar and its curve have different tangents in  $p$  in the given situation. And for that case of different tangents, the multiplicity is the product of the Orders. We know that the order of  $C$  in  $p$  is one, since it is less than the intersection multiplicity with the tangent. So lastly one needs to show, that the order of the polar in  $p$  is one.  $\square$

If we are interested in tangents, the next important question one can ask is whether a curve has inflection points. Clearly, we need to get the second derivative to study them and to get a curve from that we take the determinant of it:

**Definition 18** (Hessian curve). for a given Curve  $C = V(f)$ , we define the Hessian curve as follows:

$$H_f = \det \left( \left( \frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{0 \leq i, j \leq 2} \right)$$

is the Hessian matrix. The Hessian curve  $H(C)$  is then given by its determinant leaving us with  $H(V(f)) = V(H_f)$ .

**Remark 16.** It is clear, that this construction gives us inflection points, if we intersect it with  $C$ , and that the degree of  $H_f$  is  $3(n - 2)$ , if it's not the 0 polynomial. If  $\text{mult}_p(C \cap H(C)) = 1$ , we call it a simple inflection point, as seen in figure 12. With the theorem of Bézout, we get maximal  $3n(n - 2)$  inflection points for a curve of degree  $n$ . The simple inflection points of  $C$  correspond to simple cusps of  $C^*$ . Fischer shows, that by using the theory of numerical invariants, that are coefficients of local parametrizations,

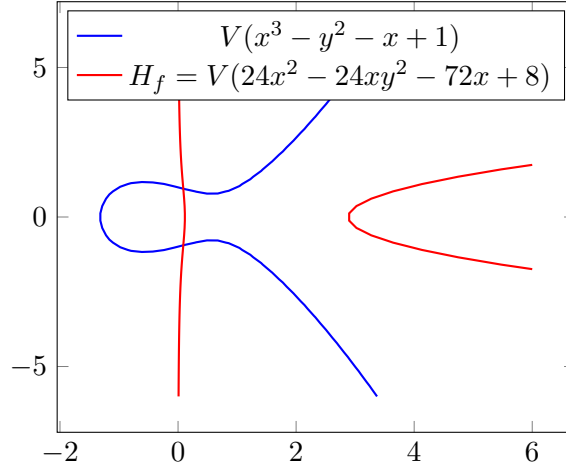


Figure 12: Hessian of an elliptic curve

As one might expect, the amount of double points limits the amount of cusps, as they seem like degenerated double points, and all of them are limited by the degree. Being a double point  $p^*$  of  $C^*$  means, that we have two distinct tangents of  $C^*$  in  $p^*$ . So  $p^*$  came from a tangent of  $C$ , and the two tangents came from two different points. Therefore, double points of  $C$  correspond to double tangents of  $C^*$ , and since  $(C^*)^* = C$ , we can say that double tangents of  $C$  correspond to double points of  $C^*$ .

Now we want to understand how the intersection multiplicity with its polar reflects simple double points and cusps.

**Lemma 4** (Intersection multiplicities with the polar and the Hessian). *Let  $C$  be a curve of higher degree than 1, where  $C$  and  $C^*$  have at most simple double points and simple cusps. Let  $p$  be a singular point of  $C$ , and  $q$  be point in  $\mathbb{P}^2$ , such that  $q$  does not lie on the Tangent of any singular point. Then*

$$\text{mult}_p(C \cap \Delta_q f) = \begin{cases} 2 & \text{if } p \text{ is a simple double point,} \\ 3 & \text{if } p \text{ is a simple cusp.} \end{cases}$$

Furthermore,

$$\text{mult}_p(C \cap H(C)) = \begin{cases} 6 & \text{if } p \text{ is a simple double point,} \\ 8 & \text{if } p \text{ is a simple cusp.} \end{cases}$$

*Proof.* One can prove this lemma either by heavy computations of the resultants or by calculating local parametrizations, which one can insert into  $\Delta_p f$  or  $H(C)$ . This way can be seen in [Fis94, p. 68]. We will omit this computational proof and look at examples instead.  $\square$

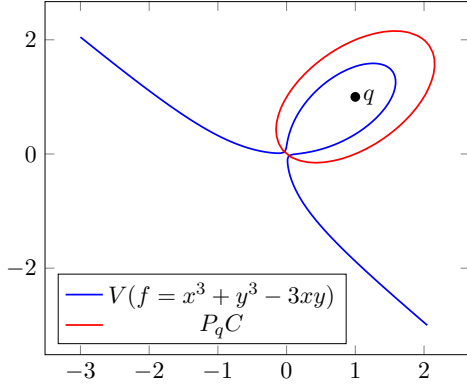


Figure 13: Intersection in a double Point

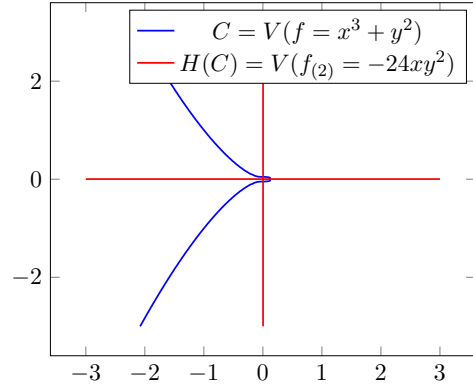


Figure 14: Hessian in a Cusp

**Example 5.** If we look at the Folium of Descartes  $C = V(x^3 + y^3 - 3xyz)$ , we get the polar  $\Delta_q f = 3x^2 - 3xy - 3xz + 3y^2 - 3yz$  in  $q = [1 : 1 : 1]$ , as seen in figure 13. We get a resultant of  $81x_1^2x_2^2$ , giving us an intersection multiplicity of 2.

If we consider the cusp  $V(x^3 + y^2z)$ , we get the Hessian curve  $H(C) = V(-24xy^2)$ , which is not given by a minimal polynomial. However, the resultant of  $x^3 + y^2z$  and  $-24xy^2$  is  $13824y^8z$ , giving is an intersection multiplicity of 8 in  $[0 : 0 : 1]$ . Those computations are easy to implement in any program that can calculate multivariable taylor expansions, such as Matlab.

**Theorem 5.** (*Plücker formula*) For a given irreducible curve  $C$  of higher degree than 1, where  $C$  and  $C^*$  have at most simple double points and simple cusps, the invariants :

$$\begin{aligned} n &= \text{degree of } C, & n^* &= \text{degree of } C^*, \\ d &= \# \text{ double points of } C, & d^* &= \# \text{ double points of } C^*, \\ s &= \# \text{ Cusps of } C, & s^* &= \# \text{ cusps of } C^*, \end{aligned}$$

have the following relations:

$$n^* = n(n-1) - 2d - 3s, \quad (4)$$

$$s^* = 3n(n-2) - 6d - 8s. \quad (5)$$

*Proof.* To prove part 4, we first choose a point  $q$  outside the finite simple double- or inflection tangents, such that there are exactly  $n^*$  tangents through  $q$  and  $p_1, \dots, p_{n^*}$  on  $C$ . By remark 15 we know that the  $\text{mult}_{p_i}(C \cap P_q C) = 1$  for all  $i \in \{1, \dots, n^*\}$ . So by the

theorem of Bézout we get

$$\begin{aligned} n(n-1) &= \sum_{p \in C \cap P_q C} \text{mult}_p(C \cap P_q C) \\ &= n^* + \sum_{p \in \text{Sing} C} \text{mult}_p(C \cap P_q C). \end{aligned}$$

By lemma 4 we have the desired formulas.  $\square$

**Remark 17** (Dual Plücker formulas). For the same situation as in theorem 5 we know that double points of  $C^*$  correspond to double tangents, and simple cusps of  $C^*$  correspond to simple inflection points, we can apply the Plücker formulas to the dual curve and get:

$$n = n^*(n^* - 1) - 2d^* - 3s^* \quad (6)$$

$$s = 3n^*(n^* - 2) - 6d^* - 8s^*, \quad (7)$$

giving us a formula to compute the amount of double Tangents.

With those formulas we have proven, that every smooth Quartic with only simple double points of its dual, has 28 bitangents, since: by the first Plücker formulas we get:  $n = 4$ , therefore  $n^* = 12$ . With the second formula, we get that  $s^* = 24$ . Inserting that into the third, we get

$$\begin{aligned} 4 &= 12 \cdot 11 - 2d^* - 3 \cdot 24 \\ 2d^* &= 12 \cdot 11 - 3 \cdot 24 - 4 = 56, \end{aligned}$$

leaving us with the desired result of 28 bitangents. For a more general theorem, we need to distinguish between hyperflexes and bitangents of a curve  $C$ . We call a line a bitangent if it is a tangent in two different points. That means, that the intersection multiplicity in 2 points is greater than 1. For a working formula, we need the limit of this, where the two points come infinitely close. So we call a line hyperflex of  $C$ , iff it intersects  $C$  with a multiplicity greater than 4.

**Lemma 5** (Contact lines are hyperflexes or bitangents). *Let  $C$  be a smooth quartic curve in  $\mathbb{P}^2$ , and let  $l$  be a line, such that  $\text{mult}_p(l \cap C) > 1$  for all intersections  $p$ . We call such a line a contact line. Then  $l$  is a hyperflex or a bitangent.*

*Proof.* By Bézout's theorem, we have

$$4 \cdot 1 = \sum_{p \in l \cap C} \text{mult}_p(l \cap C),$$

where the summands are even. So the only possible way will be a double tangent or a hyperflex.  $\square$

**Theorem 6** (28 Bitangents and Hyperflexes). *For a smooth quartic, there are always 28 contact lines.*

*Proof.* The proof can be read in [Dol12] Chapter 6.  $\square$

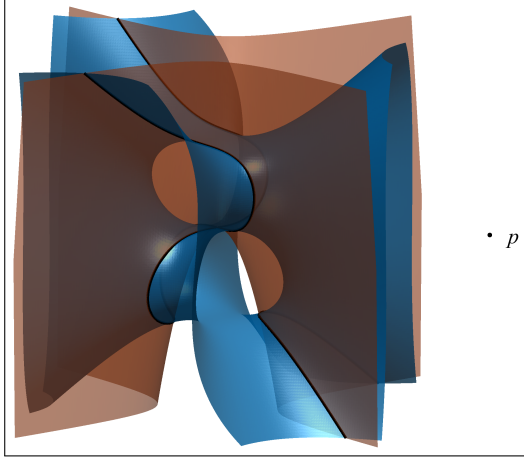


Figure 15: The polar of  $X = V(z^3 - z - x^3 + 3xy^2)$ , from  $p$  which lies outside of  $X$ .

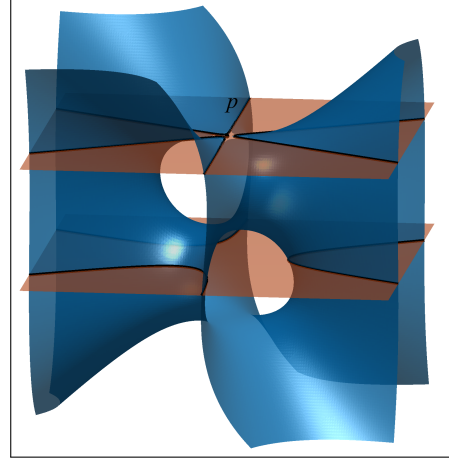


Figure 16: The polar of  $X = V(z^3 - z - x^3 + 3xy^2)$ , from  $p$  which lies on a line.

## 6 The Shadow of a Cubic

As the numbers already suggest, there is a connection between the 27 lines and the 28 bitangents, that dictates a connection between smooth cubics in  $\mathbb{P}^3$  and smooth quartics in  $\mathbb{P}^2$ . In fact, that connection can be expressed by a mapping from the smooth cubics in  $\mathbb{P}^3$  to the smooth quartics in  $\mathbb{P}^2$ , where the 27 lines of the cubic get mapped to 27 bitangents. This mapping, called a shadow, is what we will explore in detail in the next chapter. The aim however is to develop an algorithm that can calculate the shadow. Whilst the pictures will be in  $\mathbb{P}^n$ , or rather  $\mathbb{R}^3$  projected to the dimensional screen, we will develop the algorithm to work over arbitrary fields of characteristic 0. For the theory to work, we also need our fields to be algebraically closed. We will follow a construction shown in [Gro20].

**Definition 19** (Shadow of a hypersurface). For a given hypersurface  $X \subset \mathbb{P}^n$ , and a hyperplane  $H$ , and a point  $p$ , we define the shadow as follows:

$$Sh(X, H, P) = \pi(X \cap P_p X \setminus \{p\}),$$

where  $\pi$  denotes the central projection, that maps any  $q$  onto  $pq \cap H$ ,  $pq$  is the line through  $p$  and  $q$ , and  $P_p X$  is the polar given by the formal derivative.

**Remark 18.** As we have seen before, for each point  $p \in X \cap P_p$  the line through  $p$  and  $q$  is a tangent of  $X$  in  $q$ . This gives an intuitive interpretation of the shadow. As one can imaging from figure 15, this shadow would not need to be smooth. In fact, one can see that projecting the intersection of the curve and its polar, in figure 15, would give a double point. To get rid of that, we need to choose a point from the surface itself, which does not lie on one of the lines. Otherwise, we get a polar as seen in figure 16, where the

projection of the intersection either misses a lot, or contains lines as components. By Bézout, this results again in self intersections in the form of double points. So we need to choose a point on  $X$  that is not on any of the lines. To calculate that, we need to figure out, how to calculate the central projection. The trick will be, to make a change of coordinate, such that  $P = [0 : 0 : 0 : 1]$  is at infinity and  $H = V(x_3)$ . In the perspective of the affine world, we then have a parallel projection onto  $V(x_3)$  instead of a central projection and to calculate that is trivial, since we can just set the last coordinate to be 0.

So the following constructive proof will have 4 steps:

1. Calculate the transformation of coordinates, such that  $P = [0 : 0 : 0 : 1]$  and  $H = V(x_3)$ .
2. Calculate  $C = X \cap P_{[0:0:0:1]}X$ .
3. Calculate the projection onto  $H$ .
4. If wanted, then undo the transformation of coordinates.

**Lemma 6** (Change of coordinates). *Let  $X = V(f)$  be a cubic hypersurface in  $\mathbb{P}^3$ ,  $p$  be a point in  $X$  and  $H$  be a hyperplane not containing  $p$ . Then there exists a matrix  $g$ , such that  $g([0 : 0 : 0 : 1]) = p$ , and  $g(V(x_3)) = H$ .*

*Proof.* In theory, we need 3 non-collinear points  $A, B, C$  in  $H$ . The matrix  $G = [A \ B \ C \ p]$  will do the desired, since  $g([0 : 0 : 0 : 1]) = G \cdot [0 : 0 : 0 : 1]^t = p$ , and  $g(V(x_3)) = g([x : y : z : 0]) = \text{span}(A, B, C)$ . And clearly, since  $p$  is not in  $H$ , the  $A, B, C, p$  will be linearly independent and therefore  $G$  will be invertible.  $\square$

**Algorithm 1** (Calculation of the change of coordinates). To calculate  $G$ , we will do as follows:

To get there, we change the implicit form of  $H = V(f)$  to a coordinate form. We do this by first solving  $f$  for the last *non-zero* monomial in the variable  $x_i$ , so we have  $\alpha_0 x_0 + \dots = \alpha_i x_i$ . Now we define  $P_j$  to have as the  $j$ th coordinate  $\alpha_i$ , and the  $i - th$  coordinate  $-\alpha_j$ , for  $j < i$ . And for  $j < i$ , we can just define the  $j - th$  coordinate to be 1. This construction works as long as there are two monomials in  $f$ . If  $f = x_i$  is generated by a single monomial, we can just define the  $P_j$  as 1 in the  $j - th$  coordinate and for  $j = i$  we can define the 4th coordinate to be 1. We will afterwards define the last column to be  $p$  and for that reason we can just take the  $4 \times 4$  identity matrix and switch the 4th with the  $i$ -th row.

**Remark 19.** This algorithm has been programmed using Oscar and can be seen in the source code 1. The reason for Oscar is the interaction of different programs, such as singular for algebraic calculations and polymake for tropical ones.

**Lemma 7** (Intersection with the polar). *Let  $X = V(f)$  be a smooth cubic in  $\mathbb{P}^3$ , such that  $p = [0 : 0 : 0 : 1]$  is in  $X$ , and no line in  $X$  passes through  $p$ . Since  $f([0 : 0 : 0 : 1]) = 0$ , we know that  $x_3^3$  is no summand of  $f$ , and therefore  $f$  is of the form:*

$$f = x_3^2 f_1 + x_3 f_2 + f_3,$$

where the  $f_i$  are homogeneous of degree  $i$ . For those  $f_i$ , the intersection of  $X$  with its polar has the form:

$$X \cap P_p X = V(\underbrace{x_3^2 f_1 + x_3 f_2 + f_3}_f, 2x_3 f_1 + f_2).$$

Furthermore, the projection from  $p$  to  $H$ , can be restricted to an isomorphism  $\pi : (X \cap P_p X) \setminus \{p\} \rightarrow V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$ .

*Proof.* Since only  $p_3 \neq 0$ , we can calculate the polar as:

$$\Delta_p f = \sum_{i=0}^3 p_i \frac{\partial f}{\partial x_i} = 2x_3 f_1 + f_2.$$

This already proves the first part. For the second part, we realize, that the projection  $\pi : \mathbb{P}^3 \rightarrow H$  is nothing but an elimination of the last coordinate. To see that we realize, that  $p \cap H = \emptyset$ , but  $\text{span}(p, H) = \mathbb{P}^3$ . This results in the fact that  $K^4 = \mathfrak{p} \oplus \mathfrak{h}$ , where  $\mathfrak{p}$  and  $\mathfrak{h}$  denote the affine subsets of  $K^4$  belonging to  $p$  and  $H$ . Since

$$\phi : K^4 \rightarrow \mathfrak{h}, v = v_p + v_h \mapsto v_h$$

is linear, it is given by a matrix  $B \in K^{4 \times 4}$  of Rank 3 =  $\dim \mathfrak{h}$ . For a given  $[v] = [v_p + v_h] \in \mathbb{P}^3 \setminus \{p\}$ , we have that  $v_h \neq 0$ , since  $[v] \neq p$ . Now by linear algebra we have:

$$(\mathfrak{p} + \lambda \cdot v) \cap \mathfrak{h} = (\mathfrak{p} + \lambda \cdot v_h) \cap \mathfrak{h} = \lambda v_h.$$

This results in  $\pi(q) = \text{span}(p, q) \cap H = [q_h]$ , where  $q_h$  is just  $q$  with a last coordinate of 0. So if we identify  $H$  with  $\mathbb{P}^2$ , by deleting the last coordinate, we get:

$$\begin{aligned} \pi : \mathbb{P}^3 \setminus \{P\} &\rightarrow \mathbb{P}^2 \\ [x_0 : x_1 : x_2 : x_3] &\mapsto [x_0 : x_1 : x_2]. \end{aligned}$$

We already know that projections are morphisms of varieties, since they are polynomial mappings. To show that  $\pi : (X \cap P_p X) \setminus \{p\} \rightarrow V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$  is an isomorphism, we need to show 2 parts: first, that this map is well-defined, meaning that the codomain is sufficient. Then we need to show that an inverse exists.

1. Let  $q = [x_0 : x_1 : x_2 : x_3]$  be in  $(X \cap P_p X) \setminus \{p\}$ . If  $f_1([x_0 : x_1 : x_2]) = 0$ , we know that  $\Delta_p f(a) = (2x_3 f_1 + f_2)(q) = f_2([x_0 : x_1 : x_2]) = 0$ , since  $q$  is in  $P_p X$ , and finally since  $q \in X$  we know that  $f(q) = 0 + 0 + f_3([x_0 : x_1 : x_2]) = 0$ . Therefore,



the last coordinate  $x_3$  doesn't matter, giving us that  $\lambda q + \mu p$  is a line in  $x$ . This however contradicts our assumption of  $p$  not lying on a line.

Thus, we know that  $f_1([x_0 : x_1 : x_2]) \neq 0$ . From this we get that

$$0 = \Delta_p f([x_0 : x_1 : x_2 : x_3]) = 2x_3 f_1([x_0 : x_1 : x_2]) + f_2([x_0 : x_1 : x_2]),$$

which we can solve for  $x_3$ , to give us:

$$x_3 = -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}. \quad (8)$$

This means that  $0 = f(q) = (x_3^2 f_1 + x_3 f_2 + f_3)(q) = (\frac{f_2^2}{4f_1^2} - \frac{f_2^2}{2f_1} + f_3)(q) = (-\frac{f_2^2}{4f_1} + f_3)(q)$  for all  $q \in (X \cap P_p X) \setminus \{p\}$ . Therefore, the image of  $\pi$  is contained in  $V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$ . The equation 8 already tells us how we can calculate the last coordinate, giving us the idea of an inverse function  $\pi^{-1}([x_0 : x_1 : x_2 : x_3]) = [x_0 : x_1 : x_2, -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}]$ .

2. Again for the inverse we need to check that the codomain is sufficient, and that it actually is the inverse. The first is just a short calculation, where we can divide by  $f_1$ . So for all  $[x_0 : x_1 : x_2] \in V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$ :

$$\begin{aligned} f(\pi^{-1}([x_0 : x_1 : x_2])) &= f([x_0 : x_1 : x_2 : -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}]) \\ &= (-\frac{f_2^2}{4f_1} + f_3)([x_0 : x_1 : x_2 : -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}]) \\ &= -\frac{f_2^2([x_0 : x_1 : x_2])}{4f_1([x_0 : x_1 : x_2])} + \frac{4f_3 f_1([x_0 : x_1 : x_2])}{4f_1([x_0 : x_1 : x_2])} = 0 \\ &= \frac{f_2^2 - 4f_1 f_3([x_0 : x_1 : x_2])}{4f_1([x_0 : x_1 : x_2])} = 0, \end{aligned}$$

and

$$\Delta_p f([x_0 : x_1 : x_2 : -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}]) = -f_2 + f_2 = 0.$$

Since  $\pi^{-1}$  never maps to  $[0 : 0 : 0 : 1]$  we have that the codomain is sufficient. To be completely rigorous, we would need to show, that  $\pi^{-1}$  is a morphism. One would do that by showing that maps of projective varieties the form  $f = X \supset U \rightarrow Y, x = [x_0 : \dots : x_n] \mapsto [\phi_1(x) : \dots, \phi_n(x)]$  for  $\phi_i \in \mathcal{O}_X(U)$  are morphisms of ringed spaces. [Gat21a][4.7 resp. 12.26]. By equation 8 we have that  $\pi^{-1}\pi([x_0 : x_1 : x_2 : x_3]) = [x_0 : x_1 : x_2 : -\frac{f_2([x_0 : x_1 : x_2])}{2f_1([x_0 : x_1 : x_2])}] = [x_0 : x_1 : x_2 : x_3]$  and it is trivial that  $\pi\pi^{-1}([x_0 : x_1 : x_2]) = [x_0 : x_1 : x_2]$ .

□

**Remark 20** (The meaning of  $V(f_1)$ ). If we have a smooth cubic  $X = V(x_3^2 f_1 + x_3 f_2 + f_3)$ , one might ask why we have to take out  $V(f_1)$ , if we look at the shadow. To understand that, we can look at the tangent space of  $X$  in  $p = [0 : 0 : 0 : 1]$ . This space will turn out to be  $V(f_1)$ .

*Proof.* First for the ease of notation we generalize the polar of a curve  $V(f)$ , for a homogeneous  $f$  of degree  $r$  in  $p = [a_0 : a_1 : \dots : a_r]$  to higher derivatives:

$$\Delta_p f^k f = \sum_{i_1, \dots, i_k \in \{0, \dots, n\}} a_{i_1} \cdots a_{i_k} \frac{\partial^k f}{\partial x_{i_1} \cdots \partial x_{i_k}},$$

for  $0 < k < r$ . If we now take two points  $p$  and  $q$  in  $\mathbb{P}^n$ , we have for  $0 \leq s < r$ :

$$\frac{1}{s!} \Delta_p f^s f(q) = \frac{1}{(r-s)!} \Delta_q^{r-s} f(p). \quad (9)$$

That is a simple result of a comparison of coefficients after two different Taylor expansions of  $f(\lambda p + \mu q)$ :

$$\begin{aligned} f(\lambda p + \mu q) &= \sum_{i=0}^r \frac{\lambda^{r-i} \mu^i}{i!} \Delta_p^i f(q), \\ f(\lambda p + \mu q) &= \sum_{i=0}^r \frac{\mu^{r-i} \lambda^i}{i!} \Delta_q^i f(p). \end{aligned}$$

With those 2 equations, we can match the coefficient by looking at the  $s$ -th summand of the first and the  $(r-s)$ -th summand of the second equation. Since  $\lambda^{r-(s)} \mu^{(s)} = \mu^{r-(r-s)} \lambda^{r-s}$  and by equality of the Taylor expansions, the summands need to coincide, giving us the desired formula 9.

We know that the line through  $p$  and  $q$  is a tangent, iff  $p \in P_q X$ . By the calculation above, and since  $p = [0 : 0 : 0 : 1]$ , we know that

$$\Delta_q^1 f(p) = \frac{1}{3-1} \Delta_p^{3-1} f(q) = \frac{1}{2} \frac{\partial^2 f}{\partial x_3^2} \Big|_q = \frac{1}{2} f_1(q).$$

This tells us that  $q$  is in the tangent space of  $X$  in  $p$  iff  $q$  lies in  $V(f_1)$ .  $\square$

**Theorem 7.** *Let  $X = V(f)$  be a smooth cubic in  $\mathbb{P}^3$ , and  $p$  be a point in  $X$  outside the 27 lines. Let  $H$  be a hyperplane not containing  $p$ . Then  $\overline{Sh}(S, H, p)$  is a smooth quartic lying in  $H$ .*

*Proof.* By lemma 6 we can find an invertible matrix  $G$ , such that  $H = g(V(x_3))$ , and  $p = g([0 : 0 : 0 : 1])$ . We can now change the coordinates and look at  $g^{-1}(X)$ , that is a variety again, since  $g^{-1}(V(f)) : \{g^{-1}(x) \in \mathbb{P}^3 | f(x) = 0\} = \{y \in \mathbb{P}^3 | f(g(y)) = 0\}$ , where  $f(g(y))$  is a polynomial, because  $g(y)$  is a polynomial function. So our new hypersurface  $X'$  is given by the evaluation of  $f$  in  $G \cdot [x_0, \dots, x_3]^t$ . For the sake of readability, let

$X = V(f)$  be in such a configuration,  $p = [0 : 0 : 0 : 1]$ , and  $H = V(x_3)$ . By lemma 7, we know that the shadow of  $X = V(x_3^2 f_1 + x_3 f_2 + f_3)$  is  $V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$  (after eliminating the last coordinate). Now we want to show that  $V(f_2^2 - 4f_2 f_3)$  is smooth. First, we will show that  $V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$  is smooth, by making use of the Jacobian criteria. To start, we reformulate:

$$\begin{aligned} f_2^2 - 4f_2 f_3 &= (2x_3 f_1 + f_2)^2 - 4f_1(x_3^2 f_1 + x_3 f_2 + f_3) = (\Delta_p f)^2 - 4f_1 f. \\ &\Leftrightarrow 4f_1 f = (\Delta_p f)^2 - (f_2^2 - 4f_1 f_3) \end{aligned} \quad (10)$$

Now let  $q = [a_0 : a_1 : a_2] \in V(f_2^2 - 4f_2 f_3) \setminus V(f_1)$ , so  $\pi^{-1}(q) \in X \cap \Delta_p f \setminus \{p\}$ . Differentiating 10 in  $\pi^{-1}q$  gives us:

$$\begin{aligned} 4f_1(\pi^{-1}(q)) \frac{\partial f}{\partial x_i} \Big|_{\pi^{-1}(q)} &= \frac{\partial (\Delta_p f)^2}{\partial x_i} \Big|_{\pi^{-1}(q)} - \frac{\partial (f_2^2 - 4f_1 f_3)}{\partial x_i} \Big|_{\pi^{-1}(q)} - 4f(\pi^{-1}(q)) \frac{\partial f_1}{\partial x_i} \Big|_{\pi^{-1}(q)} \\ &= 2P_p f(\pi^{-1}(q)) + \frac{\partial \Delta_p f}{\partial x_i} \Big|_{\pi^{-1}(q)} - \frac{\partial (f_2^2 - 4f_1 f_3)}{\partial x_i} \Big|_{\pi^{-1}(q)} - 0 \\ &= - \frac{\partial (f_2^2 - 4f_1 f_3)}{\partial x_i} \Big|_{\pi^{-1}(q)}. \end{aligned}$$

By the smoothness of  $V(f)$ , and since  $4f_1(\pi^{-1}(q)) = 4f_1(q) \neq 0$ , there is always a  $0 \leq j \leq 2$ , such that  $4f_1(\pi^{-1}(q)) \frac{\partial f}{\partial x_i} \Big|_{\pi^{-1}(q)}$  doesn't vanish and therefore  $-\frac{\partial (f_2^2 - 4f_1 f_3)}{\partial x_i} \Big|_{\pi^{-1}(q)}$  doesn't vanish either, leaving us with the smoothness of  $V(f_2^2 - 4f_1 f_3) \setminus \{f_1\}$ .

Now we want to show smoothness of  $V(f_2^2 - 4f_1 f_3) \cap V(f_1)$ . To do that let  $q = [a_0 : a_1 : a_2] \in V(f_2^2 - 4f_1 f_3) \cap V(f_1)$ . Then  $f_1(q) = 0$ , and  $0 = (f_2^2 - 4f_1 f_3)(q) = f_2(q)$ . If  $f_3$  vanished in  $q$ , then the line  $\lambda[a_0 : a_1 : a_2 : 0] + \mu p = [\lambda a_0 : \lambda a_1 : \lambda a_2 : \mu]$  would be contained in  $X$ , since:

$$f([\lambda a_0 : \lambda a_1 : \lambda a_2 : \mu]) = \mu^2 \lambda f_1(q) + \mu \lambda^2 f_2(q) + \lambda^3 f_3(q) = 0 + 0 + 0.$$

This however is a contradiction and therefore  $f_3(q) \neq 0$ . Remark 20 tells us that  $V(f_1)$  is the tangent space and by the smoothness of  $X$  we have that this is a plane, giving us a  $0 \leq j \leq 2$ , such that  $\frac{\partial f_1}{\partial x_j} \neq 0$ . This means that

$$\begin{aligned} \frac{\partial (f_2^2 - 4f_1 f_3)}{\partial x_j} \Big|_q &= 2f_2(q) \frac{\partial f_2}{\partial x_j} \Big|_q - 4f_1(q) \frac{\partial f_3}{\partial x_j} \Big|_q - 4f_3(q) \frac{\partial f_1}{\partial x_j} \\ &= -4f_3(q) \frac{\partial f_1}{\partial x_j} \neq 0. \end{aligned}$$

Therefore the entire curve  $Q = V(f_2^2 - 4f_1 f_3)$  is smooth. With the help of Bézout this implies irreducibility, since otherwise there would be a intersection, giving us a singularity. For a scheme-theoretic argument, see [Gat21b, p. 33 5.1]. The irreducibility of  $V(f_2^2 - 4f_1 f_3)$  tells us that open subsets are dense, meaning that  $\overline{V(f_2^2 - 4f_1 f_3) \setminus V(f_1)} = V(f_2^2 - 4f_1 f_3)$ . Therefore, we have a smooth quartic in  $\mathbb{P}^2$ , or in  $V(x_3)$ .

If in the beginning our  $H$  wasn't  $V(x_3)$  we need to undo the change of coordinates, giving us  $V((f_2^2 - 4f_1 f_3) \circ g^{-1}) \cap H$  as a smooth cubic in  $H$ .  $\square$

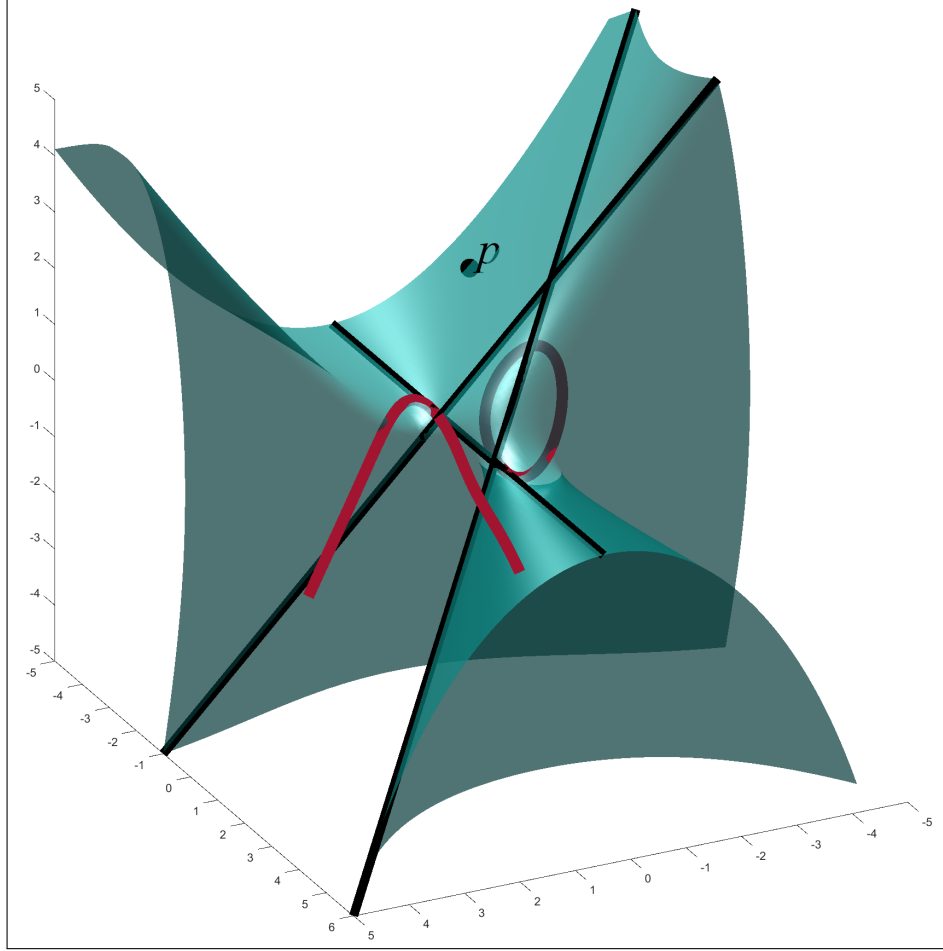


Figure 17: Shadow of a Cubic with 3 lines

**Algorithm 2** (The Shadow of a smooth cubic). To calculate the shadow of a smooth cubic, we will first do a change of coordinates as shown in the algorithm 1 and calculate the new polynomial  $f \circ g$ , by evaluating  $f$  at  $G \cdot [x_0 : \dots : x_3]^t$ . We can then calculate the  $f_i$ , by collecting those terms, where the exponent at  $x_3$  is  $3 - i$ , and evaluating those at  $[x_0 : x_1 : x_2 : 1]$ .

**Remark 21.** That algorithm can be seen in the source code 2. It also changes the ring afterwards to  $K[x_0 : x_1 : x_2]$  and produce the ideal  $\langle f_2r - 4f_1f_3 \rangle$ , such that further calculations such as tropicalization are easy to implement.

**Theorem 8** (Apparent boundary). *In the situation of theorem 7 the lines on  $X$  get mapped to the bitangents or hyperflexes of its shadow, and the last missing line is the intersection of the tangent space at  $p$  and the hypersurface  $H$ .*

*Proof.* A proof can be seen in [Bel+09, p. 156 5.8.9]. We can see that happening in figure 17.  $\square$

## 7 Conclusion and outlook

We have shown a geometric relation between cubics and quartics, and made that relation available to compute via Oscar. We have also become acquainted with the language of algebraic geometry and stated that this shadow works over an arbitrary algebraically closed field of characteristic 0. The given proofs can be justified concerning the ideal of beauty since they connected geometrical properties and analytical methods in the case of the implicit function theorem, or displayed a beautiful construction on how to think about tangents, in the case of the dual curve.

The next step is to really make use of this shadow and try to search for such a mapping in the tropical world. This step highlights the beauty of Oscar, since we can make use of Polymake and get the tropical hypersurface of a polynomial over a field with evaluation. Furthermore, we can randomly generate smooth cubics through  $P = [0 : 0 : 0 : 1]$ , and tropicalize them and finally visualize them. Afterwards we can throw the algebraically shadow and tropicalize that and then look at examples. This process can be seen in the code 5. For those cubics generated by code 3, we can see, that all tropicalized shadows look like a projection of the edges from the tropicalized cubic, onto the viewer's screen: For example we can look at

$$f = (-8 - 22t^{(22/29)} + 0(t^{(42/29)}))x^3 + (-21 - 19t^{(15/22)} + 0(t^{(35/22)}))x^2w + (23t^{(-5/4)} - 22 + 0(t^{(15/4)}))xyw + (-13t^{(-5/4)} - 15 + 0(t^{(15/4)}))xz w + (20t^{(-1/9)} - 13 + 0(t^{(19/9)}))xw^2 + (-25t^{(-3/20)} - 19 + 0(t^{(17/20)}))y^2w + (-3t^{(-21/22)} + 0(t^{(-1/22)}))yzw + (-12 - 15t^{(11/6)} + 0(t^{(31/6)}))yw^2 + (9 + 9t^{(7/5)} + 0(t^{(27/5)}))z^2w,$$

with the shadow

$$\begin{aligned} Sh = & (640t^{(-1/9)} + 25 + 1760t^{(169/261)} + 798t^{(15/22)} - 1144t^{(22/29)} + \\ & 0(t^{(349/261)}))x^4 + (-966t^{(-5/4)} - 874t^{(-25/44)} + 540 + 0(t^{(15/44)}))x^3y + \\ & (546t^{(-5/4)} + 494t^{(-25/44)} + 630 + 0(t^{(15/44)}))x^3z + (529t^{(-5/2)} - \\ & 1012t^{(-5/4)} + 1050t^{(-3/20)} + 1282 + 950t^{(117/220)} + 722t^{(15/22)} + \\ & 0(t^{(17/20)}))x^2y^2 + (-598t^{(-5/2)} - 118t^{(-5/4)} + 126t^{(-21/22)} + 114t^{(-3/11)} \\ & + 0(t^{(-1/22)}))x^2yz + (169t^{(-5/2)} + 390t^{(-5/4)} - 153 - \\ & 342t^{(15/22)} - 378t^{(7/5)} + 0(t^{(35/22)}))x^2z^2 + (-1150t^{(-7/5)} - 874t^{(-5/4)} + \\ & 0(t^{(-2/5)}))xy^3 + (-138t^{(-97/44)} + 650t^{(-7/5)} + 0(t^{(-57/44)}))xy^2z + \\ & (78t^{(-97/44)} + 0(t^{(-57/44)}))xy^2z^2 + (-234t^{(-5/4)} - 270 - 234t^{(3/20)} - \\ & 270t^{(7/5)} + 0(t^{(15/4)}))xz^3 + (625t^{(-3/10)} + 950t^{(-3/20)} + 361 + 0(t^{(7/10)}))y^4 \\ & + (150t^{(-243/220)} + 114t^{(-21/22)} + 0(t^{(-43/220)}))y^3z + \\ & (9t^{(-21/11)} + 0(t^{(-1)}))y^2z^2 + (-54t^{(-21/22)} + 0(t^{(-1/22)}))yz^3 + \\ & (81 + 162t^{(7/5)} + 81t^{(14/5)} + 0(t^{(27/5)}))z^4. \end{aligned}$$

If we tropicalize that, we get the figures 18 and 19. So the work of this thesis already gave a suggestion on how one could degenerate the mapping algebraic shadow, onto the tropical world.

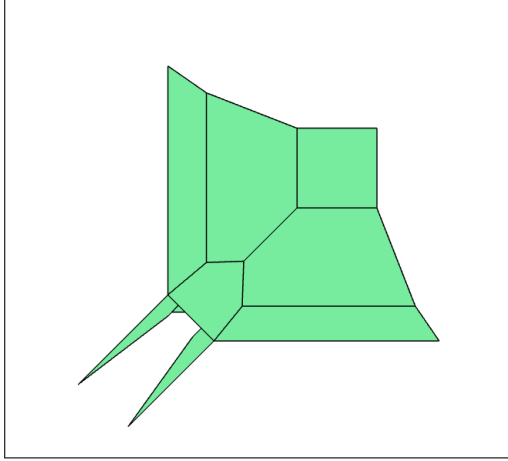


Figure 18: Tropical cubic

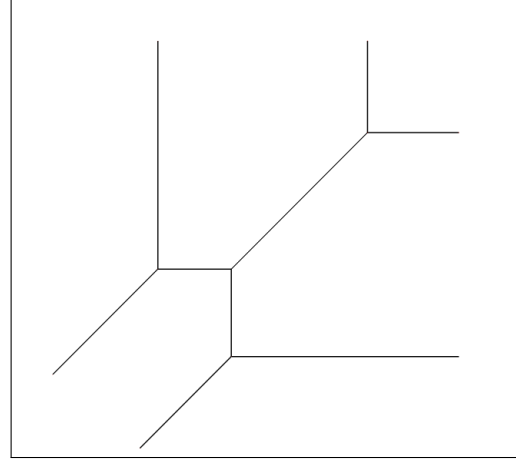


Figure 19: Tropicalized shadow

## 8 Notation

|                           |                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $V(I)$                    | The Variety corresponding to the Ideal $I$ .                                                                                             |
| $I(X)$                    | The vanishing Ideal corresponding to the Variety.                                                                                        |
| $\Delta_q^k f$            | $k$ -th polar of $f$ in $q$ (for $k = 1$ we leave out the $k$ ).                                                                         |
| $P_q X$                   | The polar variety of a hypersurface $V(f)$ given by the polar of $f$ in $q$ .                                                            |
| $\text{spec}(R)$          | The set of all prime ideals in $R$ , or the whole affine scheme associated to $R$                                                        |
| $\mathcal{O}_X(\cdot)$    | The structure sheaf on the topological space $X$ .                                                                                       |
| $\mathcal{O}_X, x$        | The stalk of $\mathcal{O}_X$ in $x$ .                                                                                                    |
| $\text{mult}_p(X \cap Y)$ | The intersection multiplicity of 2 schemes, living in a third one. If the base scheme is $\mathbb{P}^n$ , it isn't mentioned explicitly. |
| $\text{ord}_p(X)$         | The order of a point $p$ on the scheme $X$ .                                                                                             |

## References

- [BCG09] Mauro C Beltrametti, Ettore Carletti, and Dionisio Gallarati. *Lectures on curves, surfaces and projective varieties*. en. EMS Textbooks in Mathematics. Zürich, Switzerland: European Mathematical Society, Aug. 2009.
- [Bel+09] Mauro Beltrametti et al. *Lectures on Curves, Surfaces and Projective Varieties - A Classical View of Algebraic Geometry*. Zürich: European Mathematical Society, 2009. ISBN: 978-3-037-19064-7.
- [Dol12] Igor V. Dolgachev. *Classical Algebraic Geometry - A Modern View*. Cambridge: Cambridge University Press, 2012. ISBN: 978-1-139-56078-8.
- [EH16] David Eisenbud and Joe Harris. *3264 and all that: A second course in algebraic geometry*. Cambridge University Press, 2016.
- [Fis94] Gerd Fischer. *Ebene Algebraische Kurven*. de. 1994th ed. Vieweg Studium; Aufbaukurs Mathematik. Wiesbaden, Germany: Vieweg+Teubner Verlag, Jan. 1994.
- [Gat02] Andreas Gathmann. *Class notes- Algebraic Geometry*. 2002.
- [Gat21a] Andreas Gathmann. *Class notes- Algebraic Geometry*. 2021.
- [Gat21b] Andreas Gathmann. *Class notes- Plane Algebraic Curves*. 2021.
- [Gei19] Alheydis Geiger. *On realizability of lines on tropical cubic surfaces and the Brundu-Logar normal form*. 2019. DOI: 10.48550/ARXIV.1909.09391. URL: <https://arxiv.org/abs/1909.09391>.
- [Gro20] Janssen Groesbeek. *Shadow of a cubic surface*. 2020. URL: <https://studenttheses.uu.nl/handle/20.500.12932/36297>.
- [Har97] Robin Hartshorne. *Algebraic geometry*. en. 1st ed. Graduate Texts in Mathematics. New York, NY: Springer, Apr. 1997.
- [LM20] Yoav Len and Hannah Markwig. “Lifting tropical bitangents”. In: *Journal of Symbolic Computation* 96 (Jan. 2020), pp. 122–152. DOI: 10.1016/j.jsc.2019.02.015. URL: <https://doi.org/10.1016%2Fj.jsc.2019.02.015>.

## Source code

Source code 1 (The change of coordinates).

```

1  function new_coordinates(f::Union{fmpq_mpoly,PolyElem,AbstractAlgebra.Generic.MPoly},p::Array)
2  R=parent(f)
3  A=zero(identity_matrix(R,4),R);
4
5
6  if length(vars(f))>1
7  A=zero(identity_matrix(R,4),R);
8      if coeff(f,gens(R)[4]) != 0
9          A[1,1]=coeff(f,gens(R)[4]);
10         A[2,2]=coeff(f,gens(R)[4]);
11         A[3,3]=coeff(f,gens(R)[4]);
12         A[4,:]=[-coeff(f,gens(R)[1]),-coeff(f,gens(R)[2]),-coeff(f,gens(R)[3]),0];
13
14         elseif coeff(f,gens(R)[3]) !=0
15             A[1,1]=coeff(f,gens(R)[3]);
16             A[2,2]=coeff(f,gens(R)[3]);
17             A[3,:]=[-coeff(f,gens(R)[1]),-coeff(f,gens(R)[2]),0,0];
18             A[4,3]=1;
19
20             elseif coeff(f,gens(R)[2]) !=0
21                 A[1,1]=coeff(f,gens(R)[2]);
22                 A[2,:]=[-coeff(f,gens(R)[1]),0,0,0];
23                 A[3,2]=1;
24                 A[4,3]=1;
25
26             else
27                 A[1,:]=[0,0,0,0];
28                 A[2,1]=coeff(f,gens(R)[1]);
29                 A[3,2]=coeff(f,gens(R)[1]);
30                 A[4,3]=coeff(f,gens(R)[1]);
31
32             end
33     else
34         A=identity_matrix(R,4);
35         i=findall(c->c==last(vars(f)),gens(R))[1];
36         A=swap_rows!(A,i,4);
37     end
38     global A=A;
39     pb=zeros(R,4,1)
40     for i in 1:4
41         pb[i]=R(p[i]);
42     end
43     A[:,4]=pb;
44     return A
45 end
46

```

Source code 2 (The shadow of a cubic).

```

1  function Shadow_of_Cubic(I::MPolyIdeal, H::MPolyIdeal,P::Array)
2
3  # Input:

```



```

4  #I is the ideal corresponding to the cubic (so it is generated by one element)
5  #H is the ideal corresponding to the hyperplane (so the generator is of degree 1)
6  #P is the point from where the Shadow is cast (cannot be on V(H))
7
8  f= gens(I)[1] #takes the generator of I
9  h= gens(H)[1] #takes the generator of H
10 R= parent(f) #takes the polynomial Ring
11 C= base_ring(R)
12 if h(P[1],P[2],P[3],P[4])==0 #error if P lies on H
13     error("P is on the Hypersurface H")
14 end
15 if length(gens(I))>1 #error if I is generated by more then one polynmoial
16     error("I doesnt define a hypersurface")
17 end
18 if f(P[1],P[2],P[3],P[4])!=0#error if I p is not in V(I)
19     error("P is not in V(I)")
20 end
21
22 l=evaluate(f,R.(new_coordinates(h,P))*gens(R));
23
24 l_1=R.(0)
25 l_2=R.(0)
26 l_3=R.(0)
27 LCZ=[gens(R)[1],gens(R)[2],gens(R)[3],1] # eliminat the last variable
28 for i in 1 : length(l)
29     if exponent_vector(l,i)[4]==2
30         l_1=l_1+evaluate(term(l,i),R.(LCZ))
31     elseif exponent_vector(l,i)[4]==1
32         l_2=l_2+evaluate(term(l,i),R.(LCZ))
33     else
34         l_3=l_3+evaluate(term(l,i),R.(LCZ))
35     end
36 end
37 p=l_2^2-4*l_1*l_3
38 global R, (x, y, z) = PolynomialRing(C, ["x", "y", "z"]; ordering=:deglex)
39 Shadow= Oscar.ideal(R,p(x,y,z,1))
40 return Shadow
41 end

```

**Source code 3** (Generating smooth cubics going through  $P=[0:0:0:1]$ ).

```

1  function generate_cubic_throughPoint(n::Int)
2
3  A=rand(-n:n,10,3)
4  B=rand(1:n,10)
5  Q,t=PuisseuxSeriesField(QQ,20,"t")
6  R, (x, y, z,w) = PolynomialRing(Q, ["x", "y", "z", "w"]; ordering=:deglex)
7  C = map(c->f=A[c,1]+A[c,2]*t^(A[c,3]/B[c]), [1:10;])
8
9  Exp=[[1, 0, 0,2], [0, 1, 0,2], [0, 0, 1,2],[2, 0, 0,1], [1, 1, 0,1], [1, 0, 1,1], [0, 2, 0,1],
10 [0, 1, 1,1], [0, 0, 2,1],[3, 0, 0,0], [2, 1, 0,0], [2, 0, 1,0], [1, 2, 0,0], [1, 1, 1,0],
11 [1, 0, 2,0], [0, 3, 0,0], [0, 2, 1,0], [0, 1, 2,0], [0, 0, 3,0]]
12 # all partitions of 0,1,2,3 with 3 elements, where the 4th entry homogenizes
13
14     f=R(C,Exp);
15     if rank(matrix(map(i->derivative(f,i),gens(R))))==1
16         return f

```

```

17         else generate_cubic()
18     end
19
20 end
21

```

This code generates the desired polynomials. That can be seen as follows. If  $f$  is generated by this code, by inspecting the elements in `Exp` we realize, that after setting  $w = 1$  we are a polynomial in  $\langle x, y, z \rangle$ , meaning we vanish at  $(0, 0, 0, 1)$ . The smoothness is not given by the construction, but checked in line 15 since we check if the rank of the jacobian is 1. If not, we have a recursion and try again. Now this algorithm doesn't logically need to terminate, however is statistically almost always terminates. That is because the not-smooth cubics need to satisfy a polynomial equation (meaning that their derivatives vanish) giving us a space of codimension higher than 1. Therefore, almost all cubics are smooth.

**Source code 4** (Tropicalize polynomials over the Puiseux series).

```

1 function trop(f)
2     C = map(c->Rational(valuation(c)), coeffs(f))
3     E = transpose(hcat(collect(Nemo.exponent_vectors(f))...))
4     return Polymake.@pm tropical.Hypersurface{Min}(COEFFICIENTS=C, MONOMIALS=E)
5 end

```

**Source code 5** (Looking at tropical degenerations of algebraic shadows).

```

1 include("./Shadow_of_cubic.jl");
2
3 Q,t=PuisseuxSeriesField(QQ,20,"t");
4 R, (x, y, z, w) = PolynomialRing(Q, ["x", "y", "z", "w"]; ordering=:deglex);
5
6 f=generate_cubic_throughPoint(30); #The number 30 gives the range [-30:30] in which the
7                                     coefficients and exponents
8 resp the enumerator and denominator vary.
9 Tropicalf=trop(f);
10 show(Polymake.visual(Tropicalf));
11
12
13 I=Oscar.ideal(R,f);
14 H=Oscar.ideal(R,w);
15 P=[0,0,0,1];
16
17 Sh=gens(Shadow_of_Cubic(I,H,P))[1];
18 TropicalShadow=trop(Sh);
19 show(Polymake.visual(TropicalShadow));

```

## Eidesstattliche Erklärung

*Ich erkläre, dass ich die vorliegende Arbeit selbstständig angefertigt habe und nur die angegebenen Hilfsmittel benutzt habe. Alle Stellen, die dem Wortlaut oder dem Sinn nach anderen Werken, gegebenenfalls auch elektronischen Medien, entnommen sind, sind von mir durch Angabe der Quelle, als Entlehnung erkenntlich gemacht*

---

Tübingen, den 11.05.2022